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University of Zagreb
Faculty of Economics and Business
Managerial Informatics

**USER TRUST AND RISK PERCEPTION IN
CRYPTOCURRENCY TRADING PLATFORMS: AN ICT
MANAGEMENT PERSPECTIVE**

Master Thesis

Ema Hudaverdi

Zagreb, September 2025.

University of Zagreb
Faculty of Economics and Business
Managerial Informatics

**POVJERENJE KORISNIKA I PERCEPCIJA RIZIKA NA
PLATFORMAMA ZA TRGOVANJE KRIPTOVALUTAMA: ICT
MENADŽERSKA PERSPEKTIVA**

Master Thesis

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Zagreb, September 2025

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SUMMARY

This thesis examines how cryptocurrency platforms earn (or lose) user trust and how that trust translates into behaviour. Using a convergent–parallel mixed-methods design, it combines (i) documentary multiple case studies of Coinbase, Binance, and Bitstamp with (ii) a cross-sectional survey of EEA retail traders, and (iii) independent external benchmarks. The case strand assembles auditable artefacts—status histories with dated post-mortems, SOC/ISO attestations, custody architecture notes (HSM/MPC), and proof-of-reserves paired with proof-of-liabilities where available—to construct a formative “documentary trust-signals” index. The survey quantifies perceived trust, risk, self-efficacy, and behaviour (platform choice, trading intensity, precautionary settings).

Findings converge on three points. First, expectations for transparency and communication are high, while baseline trust in technical security, data protection, and fund safety is moderate and polarised. Perceived communication quality is the strongest unique correlate of whether trust actually drives platform choice. Second, self-efficacy and experience rise together with platform-directed trust; women, on average, report lower self-efficacy and higher perceived risk. Third, users distinguish exogenous “category” risks from platform-specific safeguards, rewarding concrete, checkable evidence over slogans.

Managerially, the thesis proposes a versioned Governance Assurance Pack and a Trust–Risk Capability Matrix that links real capability (controls, testing, resilience) to its legibility (clarity, cadence, and placement of evidence). Policy guidance focuses on standardising public assurance artefacts and making them machine-readable (aligned to MiCA/DORA, NIST CSF 2.0, ISO/IEC 27001). Limitations include cross-sectional self-report data and potential document survivorship bias. Future work should test which artefacts and messages causally shift trust and behaviour.

Keywords: User trust; Risk perception; ICT governance; Cryptocurrency exchanges; Proof-of-reserves.

TABLE OF CONTENT

1. INTRODUCTION.....	1
1.1. Context and Motivation.....	1
1.2. Problem Statement and Objectives.....	1
1.3. Research Questions	2
1.4 Thesis Roadmap	3
2. LITERATURE REVIEW	3
2.1. Trust theory in FinTech	3
2.2. Risk perception & behavioral finance in crypto markets	6
2.3. ICT-governance frameworks for crypto platforms	8
2.4. Critical success factors and common challenges	11
2.5. Identified research gaps.....	14
2.6. Methodological note.....	17
3. RESEARCH METHODOLOGY	18
3.1. Multiple-Case Study Design and Case Selection	19
3.2. Survey Instrument Design and Sampling Strategy.....	21
3.3. Documentary Corpus Construction and Inclusion Criteria	24
3.4. Data-Collection Procedures.....	26
3.5. Analytical Techniques	29
3.6. Limitations and Mitigation Measures.....	32
4. RESULTS.....	35
4.1. Platform Profiles and Governance Structures	35
4.2. Survey Descriptive Statistics and Correlations	36
4.3. Documentary Themes and Illustrative Evidence.....	40
4.4. Cross-Case Synthesis	41
4.5. Triangulation of Quantitative and Qualitative Findings.....	44
5. DISCUSSION	46
5.1. Interpreting Results through Trust Theory	46
5.2. Implications for ICT Governance and Management	48
5.3. Regulatory and Policy Implications	50
5.4. The Trust–Risk Capability Matrix.....	51
5.5. Study Limitations and Reliability Considerations	53
5.6. Directions for Future Research.....	54

6. CONCLUSION	56
REFERENCES	58
LIST OF FIGURES.....	62

1. INTRODUCTION

1.1. Context and Motivation

Cryptocurrency markets have matured quickly, but not evenly. The FTX collapse in 2022 and the meme-coin boom-and-bust of early 2025 reminded retail users that headlines can outrun governance: marketing promises and rapid listings often arrive long before controls, disclosure, and recourse are in place. Regulators have moved from watching to acting. In the European Union, MiCA and DORA now set clearer expectations for authorisation, safeguarding of client assets, and ICT-risk management. Exchanges have responded with a patchwork of “trust signals”—public status pages, proof-of-reserves portals, SOC/ISO attestations, and dated incident post-mortems—but their scope, freshness, and readability vary widely.

The practical problem behind this thesis is straightforward. Users say they value security and transparency, yet baseline trust remains modest and highly sensitive to incidents. ICT leaders, for their part, have no shortage of frameworks and controls; what they lack is evidence about which concrete, user-visible signals actually shift trust and behaviour for retail traders. This study tackles that gap by linking platform-side governance artefacts—such as proof-of-reserves paired with proof-of-liabilities, independent attestations with clear scope and dates, custody architecture summaries (e.g., HSM/MPC), and candid, time-stamped post-mortems—to user-side perceptions and choices.

The motivation is pragmatic. If we can show which pieces of evidence users notice, understand, and reward, platforms can invest in the right forms of legibility rather than in slogans, and regulators can encourage disclosures that truly help people judge where to trade. In short, the thesis aims to translate good ICT practice into visible, comprehensible proof—so that stronger governance becomes not only real, but reliably recognised in everyday decisions.

1.2. Problem Statement and Objectives

In retail crypto trading, trust is co-produced by organisational governance and the software artefacts that mediate use—apps, proofs, smart contracts. Yet three frictions persist. First, the visibility and quality of governance signals differ sharply across platforms: some publish clear, dated artefacts (status histories, SOC/ISO attestations, proof-of-reserves with liabilities), while others offer only slogans. Second, users vary widely in their ability to interpret technical evidence; the same solvency card or incident post-mortem can reassure an expert and confuse

a novice. Third, the field lacks integrated, time-stamped evidence that links specific ICT assurances to measurable changes in user trust and behaviour at scale. As a result, leaders have frameworks and controls, but less clarity about which signals actually move intention and use.

This thesis addresses those gaps by pursuing five tightly linked aims as a single programme of work rather than a list of tasks. It first maps the current state of user trust and perceived risk across leading centralised and decentralised venues used by EEA retail traders, establishing a baseline against which governance signals can be read. It then identifies the ICT-management levers—governance structures, security architecture, operational resilience, compliance tooling, and, crucially, communication—that most influence trusting beliefs and intentions. With that scaffolding in place, the analysis quantifies how platform-directed trust relates to behaviour, including trading frequency, retention or switching, leverage choices, and the adoption of protective settings such as MFA and withdrawal allow-lists. To turn findings into practice, the thesis develops a Trust–Risk Capability Matrix that connects two dimensions that must travel together: real capability (controls, testing, resilience) and the legibility of that capability (clear, frequent, plain-language communication). Finally, it translates the evidence into a concrete governance roadmap—an assurance pack aligned to MiCA and DORA and mapped to NIST CSF 2.0 and ISO/IEC 27001—so that strong back-office practice becomes visible, comparable proof that ordinary users can understand and act on.

1.3. Research Questions

This thesis investigates how trust is formed, maintained, and lost on cryptocurrency trading platforms by linking what firms *do* (their ICT governance and assurance artefacts) with what users *see and feel* (their perceptions, choices, and behaviours). It first asks which technical and organisational factors most shape user perceptions of trust and risk on crypto trading platforms, treating custody architecture, change and incident management, operational reliability, third-party oversight, and the clarity of communication as the core levers. It then examines how trust drivers differ between centralised exchanges and decentralised venues in the wake of high-profile failures, probing whether artefact-centric assurances (e.g., audits of smart contracts, governance timelocks, oracle and bridge disclosures) substitute for or complement institution-centric signals such as licences, SOC/ISO attestations, and status histories.

A third line of inquiry evaluates the specific contribution of compliance and transparency tools to trust formation. Here the focus is on proof-of-reserves paired with proof-of-liabilities and a named independent attester, on-chain or Merkle-tree methods, travel-rule and sanctions-screening coverage, and live status/incident dashboards with dated post-mortems, asking

whether these artefacts measurably raise trusting beliefs once their scope and cadence are made explicit. The fourth question turns to outcomes and tests how perceived risks—market volatility, technical/operational hazards, and regulatory uncertainty—shape platform choice, trading intensity, leverage use, and precautionary behaviours such as enabling multi-factor authentication, activating withdrawal allow-lists, or moving funds to cold storage.

Finally, the thesis asks what an actionable ICT-governance roadmap for European platforms should look like if the goal is to move from basic compliance to proactive trust leadership. The answer is sought in a set of concrete, testable practices—bundled as a concise “assurance pack” aligned with MiCA and DORA and mapped to NIST CSF 2.0 and ISO/IEC 27001—that make real capability legible to ordinary users and, in doing so, convert structural assurance into sustained, trust-based adoption and use. Thesis Roadmap

1.4 Thesis Roadmap

The thesis moves deliberately from motivation to evidence to action. Chapter 1 introduces the problem context, states the objectives and research questions, and explains why trust in crypto markets depends on both real ICT capability and the legibility of that capability. Chapter 2 reviews the literature on trust in FinTech (including trust in IT artefacts), risk perception and behavioural finance in crypto, ICT-governance frameworks for exchanges and DeFi front ends, and the critical success factors and pitfalls that recur, closing with research gaps that shape the study. Chapter 3 details a convergent–parallel mixed-methods design: documentary multiple case studies of Coinbase, Binance, and Bitstamp (custody/key management, change/incident management, compliance tooling, external assurance) run alongside a cross-sectional EEA survey of retail traders, with clear procedures for corpus construction, sampling, translation, reliability/validity, and GDPR. Chapter 4 reports the results from both strands and aligns them, testing whether stronger, clearer artefacts correspond to higher user-perceived trust. Chapter 5 interprets the findings through trust theory, sets out managerial and regulatory implications, and introduces the Trust–Risk Capability Matrix. Chapter 6 concludes by distilling contributions and practical guidance, noting limitations, and outlining a forward agenda. The through-line is simple: trust moves when governance is both strong and legible.

2. LITERATURE REVIEW

2.1. Trust theory in FinTech

Recent scholarship converges on a layered view of trust in FinTech. Users do not place trust only in institutions and frontline staff; they also extend it to IT artefacts—apps, models, protocols, and the data infrastructures that connect them. Devlin and colleagues (2025) make this explicit by distinguishing dispositional trust, institution-based trust, interpersonal trust, and trust in technology, and by examining how these layers interact in data-intensive AI services and decentralised finance (DeFi). Their argument is pragmatic: classic, bank-centric definitions still help, but they must be adapted because, in FinTech, the trustee can be a protocol or a model as much as a firm. That shift changes how trust is formed (through signals and proofs), how it is maintained (through governance and reliability), and how it fails (through bugs, outages, and data misuse), with effects that propagate quickly across platforms and communities (Devlin et al., 2025).

This plural view aligns with information-systems research on “human trust in IT.” Lacity et al. (2024) emphasise that users accept vulnerability toward both artefacts and the organisations that design, operate, and govern them. That dual focus reframes familiar antecedents—integrity, benevolence, competence, and communication—for software-mediated contexts. In practice, it helps explain why security controls, auditability, responsible data practices, and transparent incident handling loom so large in FinTech adoption: they translate invisible back-office safeguards into legible structural assurances that users can recognise and factor into decisions (Lacity et al., 2024).

Foundational models still provide the scaffolding once translated to FinTech. McKnight and Chervany’s (2001) typology distinguishes: (a) dispositional trust (a trait-level propensity to trust); (b) institution-based trust, comprising structural assurance (rules, guarantees, and recourse) and situational normality (the sense that “things are as they should be”); and (c) trusting beliefs and trusting intentions directed at a specific other. Two adaptations are especially salient. First is the object-of-dependence problem: in technology settings, the object is often an artefact (a wallet app, lending algorithm, or smart contract) rather than a human agent (McKnight et al., 2011). Competence and integrity therefore must be inferred from code quality, uptime, audit reports, explainability, and governance signals—not merely from staff credentials or a branch’s reputation. Second, the institutional layer matters because many FinTech services operate at the edge of evolving oversight. Users import confidence from recognisable safeguards (e.g., standardised controls, recourse processes) and category-level legitimacy (e.g., widely adopted rails, reputable counterparties). Where these are weak or opaque, perceived risk rises and trust frays—even when the interface looks polished (McKnight & Chervany, 2001).

Dispositional trust still matters, but it interacts with context. Evidence from financial services suggests that faith in humanity and a trusting stance (a calculative readiness to trust) shape trusting beliefs (Moin et al., 2015). In FinTech, those dispositions likely moderate how strongly users weight artefact-level signals—e.g., “open-source code + external audit” versus “well-known brand + app-store ratings”—when forming intentions. Interpersonal trust has not vanished either: customer-support agents, community moderators, and influencers often act as surrogate trustees at onboarding or during incidents, translating technical and policy details into understandable commitments.

Crucially, trust is not a background sentiment; it is a proximal driver of adoption and use. A large study integrating UTAUT2 with a trust-theoretic model shows that trust significantly raises both behavioural intention and actual use; service quality and reputation bolster trust, while perceived risk depresses it (Amnas et al., 2023; Venkatesh, Thong, & Xu, 2012). This fits the FinTech reality: users judge whether an app is reliable, fair, and well governed before they commit deposits, connect bank data, or delegate trading. It also clarifies why hedonic motivation contributes little in this category—financial tasks are evaluated through a utilitarian lens in which dependability dominates (Venkatesh et al., 2012; Amnas et al., 2023).

Measurement deserves careful treatment. Much of the older literature measured trustworthiness (attributes of the trustee) rather than trust as a relational state that depends on the trustor and context. Newer composite approaches in financial services argue for instruments that integrate all three facets—trustee—trustor—context—and travel across channels and technologies (Moin, Devlin, & McKechnie, 2021). For FinTech, that means operationalising trust to capture (a) artefact-level signals (reliability, security, explainability), (b) institutional assurances (rules, recourse, audits), and (c) interpersonal touchpoints (support quality, clarity of communications), then testing how their weights shift across verticals (payments vs. robo-advice vs. DeFi).

Finally, failure modes differ in speed and visibility. In branch banking, trust often eroded slowly; in FinTech, a single outage, exploit, or data-handling scandal can trigger rapid cascades via social channels. Devlin et al. (2025) therefore call for research that traces cross-layer spillovers—for example, how an artefact failure (a smart-contract bug) degrades institutional trust in the platform, or how strong institutional assurances (transparent incident response, compensation, third-party attestations) cushion interpersonal and artefact-level trust after a shock. This agenda links governance choices to user psychology and, ultimately, to sustained adoption, and it motivates the constructs used later in this thesis (Devlin et al., 2025; Lacity et al., 2024; McKnight & Chervany, 2001; Amnas et al., 2023).

2.2. Risk perception & behavioral finance in crypto markets

Risk in crypto rarely arrives neatly packaged. People feel it in price screens, in the reliability of the tools they use, and in the background hum of regulation, and those strands tug on one another. A sharp move in perpetuals can trigger liquidations that spill into spot markets. A bridge incident can sour sentiment for weeks. A regulatory announcement can change what “normal” looks like the next morning.

Start with markets. Crypto prices can gap in minutes, and leverage makes everything louder. When funding rates flip or widen, traders face margin calls. Forced unwinds push prices further, and others take their cues from that flow rather than from fundamentals. Herding, reading the crowd through prints, heat maps, or a viral thread, shows up quickly in this setting (Almeida et al., 2023; Cary, 2024; Gemayel, 2024). Familiar biases travel with it. Overconfidence stretches winning streaks until it does not. Loss and ambiguity aversion delay exits and amplify the scramble once losses feel real. The disposition effect helps momentum on the way up and accelerates the break on the way down (Almeida et al., 2023). Microstructure adds friction precisely when nerves fray: thin books, fragmented pools, and on-chain gas auctions widen spreads and degrade execution, so risk feels harder to price just when that price matters most (Bank for International Settlements [BIS], 2022).

Technical and operational risks are a different kind of noise—quieter on good days, decisive on bad ones. Code and key-management choices become investment risks the moment they fail. Priority-gas auctions and miner/maximal extractable value (MEV) can reorder transactions so a swap that looked fair at submission is sandwiched moments later; repeated experiences like that chip away at perceived fairness (Daian et al., 2019). Bridges have been a recurring single point of failure: design mistakes and operational slips have produced outsized losses, which keeps tail risk salient even in calm markets (Belenkov et al., 2025; ACM Digital Library, 2024). Oracles sit in the same neighbourhood. Newer approaches—medianization, multi-source aggregation, commit–reveal—aim to harden feeds against manipulation, but their effectiveness is not easy for typical users to judge (Cong et al., 2025; Ormer, 2025). In DeFi, where the “trustee” is often a contract and a set of interfaces rather than a human organisation, people infer competence and integrity from signals the artefact gives off: uptime, the way upgrades are governed, audit scope, and the clarity of explanations (Lacity et al., 2024). What lowers perceived technical risk are concrete, checkable assurances presented plainly: independent audits or formal verification; an active, scoped bug bounty with visible outcomes; governance timelocks and emergency circuit breakers; diversified, well-documented oracles; transparent

bridge design and oversight; proof-of-reserves paired with proof-of-liabilities for custodians; segregation of client assets; MPC/HSM-backed key custody; and incident runbooks that culminate in dated public post-mortems. When those pieces are easy to find and clearly time-stamped, users are more willing to keep funds and activity in place (Lacity et al., 2024).

The third strand is regulatory. The rulebook is still settling, but it is no longer a blank page. IOSCO’s 2023 work on DeFi stresses identifying responsible persons, managing conflicts, and using plain-language risk disclosure, while the Financial Stability Board highlights the hazards created by multi-function intermediaries and calls for consistent oversight across borders (IOSCO, 2023; FSB, 2023a, 2023b). In the EU, MiCA is being phased in—stablecoin rules arrived in mid-2024, with the wider regime applying through late-2024 and early-2025 (Autorité des marchés financiers [AMF], 2024; European Securities and Markets Authority [ESMA], n.d.). The Digital Operational Resilience Act (DORA) has applied since January 2025 and brings explicit expectations around ICT-risk management, incident reporting, resilience testing, and oversight of critical third parties to most financial entities, including many crypto-asset service providers (European Insurance and Occupational Pensions Authority [EIOPA], 2025; Mayer Brown, 2025). Consistent with broader FinTech evidence, visible regulatory support and service quality lift trust, while perceived risk pulls it down; crypto intensifies both effects because financial exposure and system complexity arrive together (Amnas et al., 2023; International Monetary Fund–FSB, 2024).

One practical lesson cuts across these layers: legibility matters. Users cannot price what they cannot see. Short, standardised “risk labels” for common protocol classes—DEXs, lending, bridges, oracles—aligned with IOSCO/FSB taxonomies help people separate market, technical/operational, and counterparty risks. Pre-trade “what-if” views that surface expected slippage, liquidation ladders, or likely oracle lag turn abstractions into choices at the moment they are needed. Live resilience dashboards—current audit status, bounty depth, time-since-last-incident, oracle diversity, bridge validator composition—give a quick read on operational health. After incidents, dated timelines with remediation milestones (and a record of whether deadlines were met) restore situational normality faster than slogans. In trust-theory terms, these are structural assurances made visible; they feed trusting beliefs and intentions in exactly the way classic models predict (McKnight & Chervany, 2001; Amnas et al., 2023).

The empirical work treats the three layers as a measurement scaffold. On markets, the analysis tracks stress markers such as funding-rate dispersion and the intensity of forced liquidations. On the technical/operational side, it codes verifiable assurance signals—audits or formal verification, bounty activity, governance timelocks, oracle/bridge disclosures, and whether

proof-of-reserves is paired with liabilities. On the regulatory side, it notes visible artefacts of compliance, including MiCA passporting status and DORA-related disclosures. Formal hypotheses sit in the Introduction; the purpose here is to show why those signals should matter to users and how prior research suggests they will be interpreted.

2.3. ICT-governance frameworks for crypto platforms

In crypto markets, trust is not an abstract feeling; it is the visible outcome of governance done well. For both centralized exchanges (CEXs) and decentralized finance (DeFi) front ends, information and communications technology (ICT) governance is the mechanism that turns risk appetite into concrete behaviours users can observe—secure engineering, predictable operations, and demonstrable compliance. In the language of trust theory, this is where institution-based trust becomes tangible: structural assurances (policies, standards, controls, recourse) and situational normality (stable, predictable day-to-day service) connect back-office practice to front-stage perception (McKnight & Chervany, 2001). Contemporary information-systems work extends this logic to software-mediated contexts, noting that the “trustee” is often an IT artefact—a smart contract, wallet app, oracle, or data pipeline—as much as the organization behind it. Effective governance must therefore span both the artefact and the firm that designs, operates, and upgrades it (Lacity et al., 2024).

A credible framework starts at the top with policy and oversight. Boards (or equivalent bodies) should set clear risk appetite for cyber and operational risk, assign accountable executives, and preserve segregation of duties so that control functions can challenge commercial pressures. Version 2.0 of the NIST Cybersecurity Framework makes these expectations explicit by introducing a GOVERN function. This function requires organizations to establish, communicate, and monitor a cybersecurity risk-management strategy that steers the traditional identify–protect–detect–respond–recover activities, enabling traceability from policy to control (NIST, 2024). When platforms surface that traceability—publishing policies, named control owners, change-management commitments, and review cadences—they give non-experts a concrete anchor for their vulnerability, precisely the role structural assurances play in trust formation (McKnight & Chervany, 2001; NIST, 2024).

Beneath policy, risk management must reflect crypto-specific exposures and the socio-technical ways they fail. A living taxonomy should cover smart-contract change risk; oracle and bridge dependencies; key custody and signing processes; miner/maximal extractable value (MEV) externalities; exchange solvency and segregation of client assets; and market-integrity risks such as manipulation or unfair access. In the European Union, the Digital Operational

Resilience Act (DORA) now codifies many of these expectations. Applicable since 17 January 2025, DORA mandates ICT-risk management, major-incident classification and reporting, resilience testing, and oversight of critical third-party providers—turning good practice into requirements and encouraging firms to evidence those practices in ways supervisors and users can understand (EIOPA, 2025; ESMA, 2025).

Security engineering and DevSecOps are where preferences crystallize into code. A secure development lifecycle should include threat modelling for contracts and interfaces; automated testing (static/dynamic analysis and infrastructure-as-code scanning); build-pipeline integrity controls; hardware- or MPC-backed key management; and least-privilege administration. Many firms organise these efforts under an information security management system. ISO/IEC 27001:2022 remains the common backbone, setting requirements for establishing, operating, and improving the ISMS. Publicly accessible certificates, statements of applicability, and release notes that tie features to controls allow users to infer competence, integrity, and communication—antecedents of trust that remain relevant even when the trustee is partly an artefact (ISO, 2022; Lacity et al., 2024).

Operational resilience complements security by assuring continuity at scale. Reliability practices—error budgets, service-level objectives (SLOs), rehearsed rollback plans for contract and API changes, disaster-recovery and backup testing, defined maintenance windows, and transparent status histories—create the situational normality that lowers perceived vulnerability in everyday use. DORA’s emphasis on resilience testing and structured incident handling reinforces making these commitments auditable. Public status pages with target uptime, time-to-detect and time-to-restore benchmarks, and dated post-incident analyses that explain root causes and remediation milestones convert internal discipline into outward trust signals (EIOPA, 2025).

Because crypto platforms are assemblages of many services, third-party and supply-chain governance is pivotal. Cloud providers, RPC endpoints, oracle networks, bridge validators, KYC/AML utilities, market-data vendors, and open-source dependencies all sit on the trust path. Good practice requires due diligence, contractual controls, continuous monitoring, and exit plans. NIST CSF 2.0 elevates supply-chain risk within the GOVERN function, and EU cybersecurity policy (including NIS2 and related technical guidance) shows what acceptable evidence looks like—mappings of requirements, example artefacts, and audit pointers. A simple but powerful cue is a public register of critical dependencies with a summary of oversight applied to each; it acknowledges that “decentralized” does not mean “dependency-free” (ENISA, 2025; NIST, 2024).

Market-integrity and conduct controls sit alongside operational concerns. For CEXs, surveillance to detect wash trading and manipulation, conflict-of-interest policies for listings and market-making, and fair-access provisions reduce information asymmetries. DeFi venues face analogous challenges in artefact form: MEV-aware order-flow designs, oracle diversity and sanity checks, rate limits, and emergency circuit breakers. International bodies have warned that multifunction crypto-asset intermediaries concentrate conflicts of interest and single points of failure; governance should separate duties, limit cross-subsidies, and disclose interlocks so that users can contextualize claims of neutrality (FSB, 2023).

Financial-crime controls and data governance complete the core. Platforms that intermediate value must implement the FATF standards for virtual assets and service providers—Travel Rule interoperability, sanctions screening, and safeguarding of customer assets—while privacy-respecting data-governance (data minimization, retention discipline, role-based access) ensures analytics-heavy models do not over-collect or over-retain personal data relative to stated purposes. Regular public updates that summarise coverage (for example, the share of counterparties reachable under the Travel Rule and changes to retention schedules) help users calibrate trust against policy baselines (FATF, 2023; FATF, 2025).

DeFi adds governance questions that must be made legible. Upgradability and administrator powers should be documented in plain language, with timelocks, multisignature thresholds, and emergency pause conditions specified up front. Oracles and bridges should be selected against clear criteria and rotated under known rules. Community voting needs transparent quorum, delegation, and proposal life cycles; treasuries require dual control and audit trails. IOSCO's DeFi recommendations call for identifying responsible persons, managing conflicts, and providing plain-language risk disclosures. Protocol teams and front-end operators can align to these expectations by publishing a minimal governance assurance pack—upgrade powers, code-audit history, bug-bounty scope and activity, oracle and bridge choices, and an incident-response playbook with target disclosure timelines (IOSCO, 2023a, 2023b).

Regulatory perimeters increasingly reward this transparency. In the EU, the Markets in Crypto-Assets Regulation (MiCA) sets out licensing and authorization for issuers and service providers and allows a transitional phase during which pre-existing firms operate under national regimes before full convergence. Platforms that explain where they stand on that path—authorization status, applicable national law, complaint routes, and safeguards for client assets—reduce ambiguity and lend substance to claims of compliance. In parallel, DORA's application since January 2025 makes operational resilience a baseline expectation for many entities that touch

crypto assets, further encouraging public evidence of risk management, incident reporting, and testing (ESMA, 2023; ESMA, n.d.; EIOPA, 2025; ESMA, 2025).

Ultimately, governance only builds trust if it is legible. High-performing platforms convert internal controls into user-facing proofs: signed release notes and planned change windows for contract upgrades; status pages with historical uptime and dated post-mortems; current third-party security reports (and the hash-anchored artefacts they rely on); active bug bounties with disclosed scope and outcomes; segregated-asset attestations and proof-of-reserves paired with proof-of-liabilities (ideally under independent assurance); and plain statements of admin powers, emergency procedures, and lines of recourse. Each element maps to core antecedents of trust—competence (“we can deliver”), integrity (“we do what we say”), and communication (“we explain this clearly”)—and, crucially, makes the object of dependence explicit, whether that object is a firm, a contract, or the interface that binds them (McKnight & Chervany, 2001; Lacity et al., 2024).

Certain anti-patterns reliably corrode trust. Security by obscurity—no audits, undisclosed admin powers, or surprise upgrades—signals weak structural assurance. Solvency “proofs” that omit liabilities or lack independent assurance invite doubt. Multifunction intermediaries that combine exchange, brokerage, market-making, custody, and token issuance without clear governance separation undermine situational normality by design. Frameworks that acknowledge these pitfalls and respond with public evidence do more than satisfy regulators; they lower perceived risk for ordinary users, which is exactly the point of institution-based trust in high-ambiguity, high-stakes markets (FSB, 2023; McKnight & Chervany, 2001).

2.4.Critical success factors and common challenges

Across the FinTech literature, the drivers of initial adoption and continued use show a striking regularity. People adopt when a service clearly saves time, effort, or money and remains easy to learn; in UTAUT2 terms, performance expectancy and effort expectancy dominate early decisions. Continued use depends on the social and infrastructural context: social proof lowers the first-use barrier, habits form around convenient routines, and the surrounding infrastructure removes friction; these are the familiar roles of social influence, habit, and facilitating conditions (Amnas et al., 2023). Layered over these technology factors, trust acts as the multiplier that turns one-off trials into durable engagement. High service quality, credible reputation, and visible regulatory support raise trust, while perceived risk depresses it (Amnas

et al., 2023; McKnight & Chervany, 2001; Sekhon et al., 2014). Consistent with a utilitarian reading of financial tools, a large UTAUT2–TTM study found that hedonic motivation did not significantly predict intention, reinforcing the idea that reliability and fairness matter more than “fun” in this category (Amnas et al., 2023).

Within this frame, several capabilities consistently distinguish resilient platforms. Operational reliability is the lived face of competence: clear service-level objectives, rehearsed rollback and incident procedures, and public status histories make day-to-day performance predictable and lower users’ sense of vulnerability—what trust theory would call situational normality (McKnight & Chervany, 2001). Security, in turn, builds credibility only when it is demonstrably governed. Secure development practices, independent audits, active bug bounties, prudent key management, and timely, candid post-mortems shift integrity from marketing language into evidence (Lacity et al., 2024). For custodial models, solvency and asset-safety signalling must be truthful and complete: proof-of-reserves is persuasive when paired with proof-of-liabilities, segregation disclosures, and third-party attestations, because these artefacts can be checked rather than merely believed. Friction-managed user journeys—from onboarding and funding to recovery and dispute resolution—translate “facilitating conditions” into felt convenience, reducing the cognitive cost of safe behaviour. Finally, governance clarity closes the loop between organisation and artefact: users need to know who can change what, under which checks, and with what recourse, so that institution-based trust (rules, guarantees, and recourse) connects cleanly to trust in the software they actually touch (Devlin et al., 2025; Lacity et al., 2024).

Communication gives these capabilities most of their practical impact. Users cannot evaluate complex risks if platforms speak only in technical or legal code. The teams that succeed pair expert-legible artefacts—audit letters, change logs, and security.txt—with plain-language explanations that separate market risk from technical/operational and counterparty risks, set expectations about failure modes, and explain who bears which risks under which contingencies. This dual register maps directly to antecedents of trust—competence, integrity, benevolence, and communication—and helps convert structural assurances into trusting beliefs and, ultimately, intentions to use (McKnight & Chervany, 2001; Sekhon et al., 2014). In short, strong controls without accessible explanation remain invisible; accessible explanation without strong controls rings hollow. Trust grows when both travel together.

Alongside these success factors, several recurring pitfalls erode adoption and continuance. “Trust theatre” is the most common: surface-level badges or one-off proofs that omit liabilities, lack scope, or go stale raise perceived risk rather than reducing it. Over-indexing on novelty or

gamification is another trap. Because hedonic motivation has a weak role in FinTech intentions, entertaining mechanics rarely compensate for gaps in reliability or clarity and often correlate with short-lived activation and poor cohort retention (Amnas et al., 2023). Supply-chain opacity also undermines confidence. Modern FinTech stacks depend on vendors—cloud providers, data pipelines, oracles, and payment rails—and when those dependencies and oversight practices are not disclosed, users infer hidden single points of failure and discount the brand by association (Lacity et al., 2024). Governance ambiguity, particularly in crypto markets, has a similar effect: if administrator powers, upgrade rights, and emergency procedures are unclear or concentrated without checks, users cannot anchor their vulnerability even when the interface looks polished (Devlin et al., 2025). Finally, regulatory uncertainty continues to depress continuance intention where firms fail to translate their status into user-visible commitments—license or authorisation, routes for complaints, and safeguarding of client assets—leaving ambiguity in place despite solid product–market fit.

The managerial response that follows from this evidence is practical and testable. Reliability should be hard-wired through explicit SLOs, error budgets, and tested recovery and rollback plans, with a live status history that makes performance observable. Security and solvency evidence should be made public with current audits, clear bug-bounty scope and activity, and reserve proofs paired with liabilities coverage and named attestors. Risk communication should be designed for a phone screen: concise labels that separate types of risk and link to deeper artefacts for those who want to read further. A compact “governance assurance pack” should bring together the essentials—who can change code, which timelocks and multisig thresholds apply, which oracles or bridges are used and why, and how and when incidents are disclosed—so that users can see governance in one place rather than assemble it piecemeal. Service quality should be treated as a trust instrument in its own right: fast, informed support and closed-loop complaint handling consistently lift perceptions of integrity and benevolence (Sekhon et al., 2014). These moves are not merely good hygiene; they operate along the pathways documented in adoption research by raising perceived service quality and reputation, reducing perceived risk, and thereby increasing trust, intention, and use (Amnas et al., 2023).

Measurement closes the loop between intention and outcome. Platforms should track leading indicators of trust and habit formation, such as activation-to-first-transaction time, 30/90-day retention, depth of repeat use by product, resolution times for security or service incidents, and sentiment about the clarity of risk disclosures. They should also measure trust directly using composite instruments that capture trustee attributes, trustor disposition, and contextual safeguards rather than relying on a single proxy such as “security” (Moin, Devlin, &

McKechnie, 2021). By linking these measures to specific governance and communication practices, teams can see whether improvements in structural assurance are, in fact, moving trusting beliefs and intentions in the expected direction.

Finally, the literature points to a set of propositions that managers can evaluate in their own settings. Platforms that publish a complete governance assurance pack—combining audit history, active bug bounties, administrator-key policy, incident runbook, and proof-of-reserves with proof-of-liabilities—should exhibit higher institution-based trust and stronger continuance intention than matched peers that do not. Firms that translate regulatory status into front-end commitments—license or authorisation badges, safeguarding disclosures, and complaint routes with service levels—should experience a larger effect of perceived regulatory support on intention to use. Teams that invest in plain-language risk labels should observe lower ambiguity aversion and faster recovery of activity after incidents. Each proposition follows directly from the trust-theoretic pathway—institution-based trust leading to trusting beliefs, which shape intentions, which in turn predict use—and from the multi-study evidence on FinTech adoption (Amnas et al., 2023; Devlin et al., 2025; McKnight & Chervany, 2001).

2.5. Identified research gaps

Building on the July 2025 *Electronic Markets* agenda on trust in FinTech and the accumulating UTAUT2–TTM evidence, five closely linked gaps stand out. Together, they sketch a programme of work that treats trust not as a single attitude but as a layered, dynamic relation between users, organisations, and the IT artefacts that mediate financial activity.

The first gap concerns the object(s) of dependence in hybrid socio-technical systems. In contemporary FinTech—and especially in crypto and DeFi—the “trustee” is distributed across a brand and its governance arrangements, an app and its user experience, a model or data pipeline that recommends or decides, and a protocol or smart contract that executes. Recent work explicitly calls for unpacking this multi-object trust, noting that classic person-to-person notions only partially “travel” to artefact-centric contexts (Devlin et al., 2025). What the field lacks are models that represent trust as a multi-level construct—narrow-scope beliefs and intentions toward specific artefacts (for example, a bridge, an oracle, or a solvency-proof mechanism) nested within broader trust in the provider and the institutional environment. Those models should also capture cascade dynamics: how a protocol incident erodes artefact trust, spills into organisational reputation, and finally shapes institutional trust in the category.

Multilevel structural equation models or Bayesian hierarchical approaches that fuse on-chain telemetry (e.g., TVL, withdrawal flows) with off-chain survey measures and dated event histories are well suited to test where trust is actually “carried,” and how it moves across layers when systems fail.

A second gap is measurement. Many adoption studies still proxy “trust” with single provider attributes or generic Likert items. The next wave should adopt composite instruments that integrate the trustor (disposition to trust; trusting stance), the trustee (competence, integrity, benevolence, communication) as applied to both firms and artefacts, and contextual safeguards (structural assurances and situational normality) (McKnight & Chervany, 2001). Recent financial-services work shows that such indices are feasible and psychometrically robust (Moin, Devlin, & McKechnie, 2021). Extending these scales to artefact contexts would enable within-subject tracking of trust around governance or incident events—before/after a code upgrade, an outage, the publication of an audit, or a regulatory authorisation—to estimate how strongly each signal moves the latent construct. In parallel, behavioural proxies (for example, time-to-withdrawal after alerts, the concentration of deposits across venues, or the elasticity of repeat use to fee/latency shocks) should be calibrated against measured trust to establish criterion validity.

The third gap links concrete governance moves to psychological mechanisms. UTAUT2–TTM studies show that perceived regulatory support, service quality, and reputation raise trust while perceived risk depresses it (Amnas et al., 2023). What remains under-identified is the causal potency of specific interventions: independent security attestations and formal-verification reports; proof-of-reserves paired with proof-of-liabilities; disclosures on administrator keys and timelocks; incident-response timelines with post-mortems; or third-party benchmarks for complaints handling. Field experiments and quasi-experiments can fill this gap. For centralised platforms, staged releases of solvency attestations or enhanced incident dashboards lend themselves to difference-in-differences or synthetic-control estimators on retention and deposit flows. For DeFi protocols, randomised interface banners (e.g., “audit addendum posted,” “bug bounty increased,” “governance timelock raised”) can support A/B tests on liquidity provision, re-stake decisions, or leverage choices. Measuring perceived risk, institutional trust, and trusting beliefs alongside behaviour would reveal which governance signals most efficiently convert structural assurance into lower vulnerability and higher continuance intention.

A fourth gap concerns communication design for crypto risk. High-variance markets magnify ambiguity aversion; over-warning, however, can suppress appropriate usage. The research task is to identify risk labels, defaults, and interface cues that calibrate perception without inducing

paralysis. Conjoint experiments can test which combinations of disclosures (e.g., oracle-set composition, bridge-validator policy, emergency-pause conditions, insurance terms) users understand and value. Pre-trade “what-if” views that simulate slippage, liquidation ladders, and oracle lags can be evaluated in laboratory tasks and live A/B tests for their effects on comprehension, confidence, and conservative settings. The goal is a set of plain-language risk “nutrition labels” and human-centred defaults that lift decision quality without relying on expertise. Because communication is itself an antecedent of trusting beliefs (McKnight & Chervany, 2001), isolating its incremental contribution—over and above audit or authorisation status—would guide where teams should invest scarce attention.

The fifth gap is equity and heterogeneity. Dispositional trust, risk tolerance, and digital literacy vary by age, gender, experience, and culture; these differences likely moderate how governance and communication signals are received. Some evidence suggests younger cohorts may trust FinTech more than incumbents; however, post-incident dynamics, platform type, and local regulatory regimes can reverse that pattern. Cross-cultural, multi-group SEM and hierarchical models should therefore be standard rather than optional. Designs that intentionally oversample low-literacy or high-ambiguity-averse users can test whether expert-legible artefacts (e.g., audit hashes) and novice-intelligible cues (e.g., status pages, route-to-recourse) differentially affect perceived risk and continuance. The stakes are distributional: without careful design, risk is externalised to those least equipped to evaluate it, precisely where trust deficits are most persistent.

These gaps point to concrete methodological moves. Researchers should pair measurement innovation—composite scales that bind trustor, trustee, and context—with quasi-experimental identification around dated governance events and field experimentation in product interfaces. They should triangulate self-report, behavioural telemetry, and institutional signals (licences, audits, service-level evidence) to link governance practice to shifts in perceived risk, trusting beliefs, intentions, and use. And they should report heterogeneous treatment effects, not only averages, to inform equitable design. The resulting evidence would advance the *Electronic Markets* agenda by specifying where trust resides in hybrid systems, how it can be validly measured, and which operational and communication interventions reliably turn structural assurance into lower perceived risk and higher continuance—translating trust theory into practical levers for safer, more inclusive FinTech (Amnas et al., 2023; Devlin et al., 2025; McKnight & Chervany, 2001; Moin et al., 2021).

2.6. Methodological note

FinTech evolves on start-up time. Products ship in weeks, policy baselines shift mid-stream, and some of the most decision-relevant insights first appear in consultation papers, technical standards, incident post-mortems, or audit notes rather than in journals. In this environment, a narrative integrative review is a practical and defensible choice. Rather than attempting an exhaustive enumeration of sources, it aims to be explicit, balanced, and useful: clear about scope, open to high-quality non-academic evidence, and organised around mechanisms that matter for user trust (Devlin et al., 2025).

This approach serves three purposes. First, it legitimises mixed evidence. A regulator's MiCA or DORA memo, an ISO/IEC 27001 control requirement, and a peer-reviewed trust study can be read together to illuminate how governance signals are produced and perceived in practice. That is appropriate here because users form trust not only from theory but from rules, audits, interfaces, and status pages they actually encounter (Snyder, 2019). Second, it keeps pace. Systematic reviews are powerful but slow; a narrative synthesis can surface converging themes and live disagreements now, then indicate which questions merit immediate testing. Third, it translates evidence into action. By organising findings around causal pathways—how structural assurances lower perceived risk, how communication shapes trusting beliefs—the review generates guidance that designers, engineers, and risk leaders can implement, not merely cite.

Importantly, narrative does not mean lax. To protect rigour, the review makes its search corridors and inclusion rules explicit; privileges primary, dated artefacts where possible; and ties every claim to a traceable source. It states what is in scope (FinTech and crypto-exchange governance and user trust) and what is not (purely technical protocol optimisation without user-facing implications). It also resists over-generalisation across contexts, flagging when an insight likely travels and when it may be jurisdiction-, product-, or cohort-specific (Bell et al., 2022; Snyder, 2019).

Procedurally, the review combines scholarly databases with authoritative grey literature—standards, supervisory guidance, audit and assurance artefacts, and incident post-mortems—and records as-of dates for time-sensitive materials. Selection emphasises authenticity (official domains, named issuers/attestors), scope clarity (systems and periods covered), and relevance to the trust mechanisms developed in this thesis (institution-based trust, artefact trust, and their interfaces). Where multiple sources conflict, the review defaults to the most conservative, better-substantiated evidence and notes the discrepancy.

Finally, the review acknowledges its limits. Narrative integration cannot eliminate publication or survivorship bias, and it cannot prove causal effects. To mitigate these constraints, it triangulates across source types, signals uncertainty where appropriate, and treats its conclusions as hypotheses to be tested. The resulting synthesis does not close the inquiry; it sets the stage for the next steps outlined in this thesis—targeted experiments, quasi-experiments around dated governance events, and deeper, systematic reviews as the literature matures (Devlin et al., 2025; Snyder, 2019).

3. RESEARCH METHODOLOGY

This thesis employs a convergent–parallel mixed-methods design to examine how users form trust and perceive risk on cryptocurrency trading platforms. On the qualitative side, it assembles

documentary multiple-case studies of three contrasting exchanges—Coinbase, Binance, and Bitstamp—to trace how ICT governance is enacted in practice. On the quantitative side, it fields a cross-sectional online survey of retail traders in the EEA to capture perceptions, risk appraisals, and behaviours. A third evidentiary strand draws on independent external benchmarks (e.g., CER.live security ratings, Chainalysis risk reports, ESMA/MiCA guidance) to contextualise and triangulate both sources. The strands are analysed separately and then integrated through joint displays to assess where platform signals (audits, status pages, proof-of-reserves) align—or fail to align—with user-level trust and behaviour (Creswell & Plano Clark, 2018; Yin, 2018).

3.1. Multiple-Case Study Design and Case Selection

This study uses a multiple-case design because a single exchange, however detailed, can be idiosyncratic and difficult to generalise from. Placing three contrasting platforms side by side—Coinbase, Binance, and Bitstamp—allows the analysis to look for patterns that recur under different governance models and regulatory footprints, supporting analytic generalisation rather than anecdote (Yin, 2018). Case selection followed a maximum-variation logic grounded in prior theory and industry practice on trust. Together, the three exchanges span distinct organisational forms (a publicly listed U.S. issuer, a globally distributed private group, and a long-tenured EU-rooted exchange), divergent regulatory postures across the U.S., EU (including MiCA), and other regimes, different styles and depths of assurance signalling, and varied operational histories with respect to outages, withdrawal spikes, and incident remediation. The purpose is not to identify “winners,” but to observe how trust is operationalised through ICT governance under materially different conditions.

The unit of analysis is the platform-level ICT governance system. Within each case, the inquiry follows four embedded sub-units that are especially salient for user trust: custody and key management (design and control of wallets, private keys, and signing processes); change and incident management (how production changes are governed and how incidents are prevented, detected, and resolved); compliance tooling (e.g., transaction monitoring, sanctions screening, and market-abuse controls); and external assurance artefacts (audits, certifications, attestations, and cryptographic proofs). Making these sub-units explicit keeps the discussion concrete: instead of treating trust as an abstraction, the analysis traces it to design choices, controls, and signals that users and regulators can, in principle, observe.

Evidence is assembled as a curated documentary record rather than through interviews. The corpus includes annual and quarterly reports; security and transparency pages; proof-of-

reserves (PoR) and, where available, proof-of-liabilities (PoL) artefacts; ISO/IEC 27001 certificates and SOC reports (e.g., SOC 2 Type II) where they exist; bug-bounty policies and disclosed findings; incident post-mortems and status-page histories; technical white papers on custody and signing; regulatory filings, authorisations, and enforcement documents; and reputable third-party assessments and benchmarks. Because solvency signalling is central to contemporary trust debates, the analysis distinguishes PoR schemes that present asset holdings only from those that pair assets with explicit liabilities coverage (PoL) and an independent attester. It also records the proof method (e.g., Merkle-tree attestations versus on-chain commitments) and the scope and cadence of any assurance.

To preserve provenance and enable auditability, documents are included only if they are authentic (traceable to a verifiable source), time-bounded (with clear as-of dates), scope-specific (with unambiguous system and asset boundaries), and relevant to one of the sub-units above. Each captured item is stored with metadata—source URL, collection timestamp, and a SHA-256 hash—to create a basic chain of custody. When platforms update or replace artefacts such as reserve proofs or audit letters, superseded versions are retained alongside the new ones so that change history is visible rather than overwritten. The study explicitly time-bounds the evidence to the fieldwork window and cites as-of dates in the case narratives to help readers judge the currency of each claim.

Selecting three prominent exchanges raises comparability concerns because scale, geography, and product mix can confound governance choices. The design addresses this in two ways. First, within-case descriptions make size, jurisdiction, and business scope explicit so that like-for-like comparisons are only drawn where appropriate. Second, the cross-case synthesis is anchored in signals rather than absolute outcomes—for example, whether a platform discloses the scope of its reserve proof and the identity of any independent attester; whether status pages provide incident post-mortems with remediation commitments; and whether certification claims are backed by accessible certificates and valid report periods. External benchmarks (e.g., CER.live security ratings, Chainalysis and Coincub risk indicators, and supervisory registers) serve as a standing reference to temper platform self-descriptions.

Where non-disclosure agreements permit, the corpus may include redacted board- or risk-committee materials that illuminate governance routines or control decisions; any such material is reported only in anonymised aggregate to respect confidentiality. Ethical handling extends to the public corpus as well: all files are stored on GDPR-compliant, EU-hosted infrastructure with versioning and encrypted backups, and the data-management plan specifies retention and destruction schedules.

Taken together, these design choices balance breadth with depth. Multiple cases create room for pattern matching across heterogeneous settings; clearly defined sub-units keep the analysis tethered to concrete ICT-governance practices; disciplined documentary inclusion and chain-of-custody procedures make the evidence auditable; and explicit time-bounding guards against hindsight. The resulting case records feed the qualitative analysis in Section 3.5, where expected mechanisms from trust theory—such as structural assurances and situational normality—are tested through pattern matching and cross-case synthesis.

3.2. Survey Instrument Design and Sampling Strategy

The survey is designed to quantify perceived trust and perceived risk and to test whether visible ICT safeguards—such as two-factor authentication (2FA), withdrawal allow-lists, and proof-of-reserves dashboards—track systematically with user behaviour, including trading frequency, retention or switching, leverage use, and precautionary practices. To preserve comparability with prior research while fitting the crypto-exchange context, the instrument adapts validated scales and anchors all latent constructs on a seven-point Likert scale (1 = strongly disagree; 7 = strongly agree). Behavioural variables are captured as quantitative measures (for example, number of trades over 30/90 days, average trade size, leverage ratio) rather than Likert assessments, thereby separating attitudes from actions.

Trust is modelled as a second-order reflective construct composed of the classic trusting beliefs—competence, integrity, benevolence, and communication—and trusting intention, extended with an explicit artefact-directed trust dimension that references software-mediated signals (for example, “I trust this platform’s proof mechanism to reflect true solvency”). This follows the interdisciplinary typology of McKnight and Chervany (2001) and is specified either via the repeated-indicators approach or a two-stage approach in the structural model (see Section 3.5). Institution-based trust is operationalised as two reflective first-order constructs: structural assurances (audits, licences, reserves/liabilities evidence) and situational normality (stable operations and a predictable user experience). To capture technology-adoption antecedents, the survey includes the UTAUT2 dimensions—performance expectancy, effort expectancy, social influence, habit, and facilitating conditions—adapted verbatim where appropriate from Venkatesh, Thong, and Xu (2012). Perceived risk is partitioned into market-volatility risk, technical/operational risk (custody failure, smart-contract bugs, outages), and regulatory/legal risk. Outcome measures record behaviour over 30- and 90-day windows to reduce recall bias and include trading frequency, average trade size, product mix (spot, derivatives, DeFi), leverage use, retention/switching, and precautionary configurations (2FA

enabled, allow-list on, proportion moved to cold storage). Demographic and experience covariates cover age, gender, education, country, years trading, self-assessed risk tolerance, and digital literacy.

The questionnaire is structured around platform-specific blocks for the respondent's primary and secondary exchanges; "primary" is defined in-survey as the exchange used most frequently in the last 90 days. To minimise order effects, lists of named platforms are randomised and always include "Other (specify)" and "Not applicable." Terminology that may be unfamiliar (e.g., proof-of-reserves, withdrawal allow-list) is supported by brief tooltip glossaries to keep responses focused on perceptions rather than misunderstandings. The instrument embeds two attention checks and one instructed-response item, randomises item order within constructs where this does not disrupt coherence, and uses neutral wording. A marker variable (a short, theoretically unrelated scale such as social desirability or need-for-cognition) is included to facilitate ex-post assessment of common-method bias alongside procedural remedies. Sensitive items offer a "prefer not to say" option.

The target population is adult retail traders (18+) resident in the European Economic Area who have executed at least one crypto trade in the past 90 days. Recruitment blends reputable online research panels with country quotas (to avoid over-representation of large markets) and opt-in invitations in public communities where this is permitted by platform rules. Screening questions exclude institutional/professional participants and purely buy-and-hold users without qualifying activity. An a-priori power analysis for small effect sizes ($f^2 \approx .05$, $\alpha = .05$, power $\geq .80$) indicates that $N \approx 300\text{--}400$ is sufficient for the planned multiple-regression/PLS-SEM models; the study therefore targets ≥ 400 valid completes to enable subgroup analyses. To monitor mode effects, the survey records the recruitment source (panel vs. community) for robustness checks.

After fieldwork, post-stratification weights are calculated on country $\times$ age $\times$ gender using Eurostat distributions to correct residual imbalances from the mixed recruitment approach. Weighted and unweighted estimates are reported in the Results chapter, and substantive conclusions are checked for sensitivity to weighting. Prior to launch, the instrument undergoes a cognitive/pilot test with approximately 30 respondents to assess clarity, device behaviour, and completion time; feedback informs minor wording adjustments and item pruning. If the survey is deployed in multiple languages, items are translated and back-translated (e.g., Croatian, German) to preserve semantic equivalence, and response language is flagged to enable measurement-invariance testing.

Data integrity procedures are designed to reduce fraud and duplication. The platform applies CAPTCHA and—where legally permitted and consented—light-touch device fingerprinting. A pre-registered cleaning protocol removes speeders (e.g., completion time < one-third of the median), straight-liners/long-string responders, and multivariate outliers based on Mahalanobis distance. Behavioural outliers (for example, extreme trade counts) are winsorised at the 1st/99th percentiles for regression robustness, with raw values retained for descriptive tables. All decisions are documented and reproducible.

Measurement quality and bias are assessed using standard criteria described in Section 3.5. Reliability and validity checks include Cronbach's alpha and composite reliability (thresholds $\geq .70$), average variance extracted (AVE $\geq .50$), and discriminant validity via HTMT ($< .85$). Given multiple language versions, MICOM procedures (measurement invariance of composites) or multi-group CFA are planned before comparing groups. Common-method bias is evaluated through the marker-variable technique and sensitivity tests (Harman's single-factor and Lindell–Whitney), complementing the procedural controls noted above. Structural models use bootstrapping with 5,000 resamples; when applying PLS-SEM to ordinal indicators, robust options for consistent estimation are used; for general linear models, HC3 robust standard errors and Spearman correlations (where distributions are non-normal) are applied.

Ethics and privacy safeguards align with GDPR. The survey is anonymous and collects no direct identifiers. IP addresses are not stored; any optional contact details for prize draws, if offered, are captured in a separate database that cannot be linked back to responses. Participation is based on informed consent recorded at entry; withdrawal is possible at any time via a clearly provided contact address for data requests and deletion. Data are stored on EU-hosted infrastructure with encryption at rest and in transit, retained for 24 months, and then securely deleted according to the data-management plan. Any device-fingerprinting used for fraud prevention is applied only with explicit consent.

In sum, the instrument balances psychometric rigour with accessibility for retail traders, and the sampling strategy blends reach with safeguards for representativeness and data quality. The resulting dataset is designed to test, with appropriate validity and bias checks, whether the governance signals identified in the case studies are visible to users at scale and whether they are associated with measurable differences in trust and behaviour (McKnight & Chervany, 2001; Venkatesh, Thong, & Xu, 2012).

3.3. Documentary Corpus Construction and Inclusion Criteria

This study privileges documents over interviews because, in cryptocurrency markets, artefacts carry the strongest and most verifiable claims. Audit opinions, proof-of-reserves (PoR) and proof-of-liabilities (PoL) disclosures, status pages, incident post-mortems, certifications, and supervisory filings are the places where platforms commit to specifics—scope, dates, controls, and responsibilities. Read comparatively across cases, these materials render ICT governance auditable, time-stamped, and comparable.

The corpus is time-bounded to artefacts with explicit as-of dates between 1 September 2024 and 31 March 2025. Later corrections or replacements are also retained, versioned alongside the originals to preserve the historical trail. Analyses always cite the as-of date used. This window balances recency with enough elapsed time to observe governance cycles such as annual audits and quarter-by-quarter reserve attestations.

Inclusion follows four simple rules. A document enters the corpus only if it is authentic (retrieved from an official exchange domain, a regulator’s register, or a named attestor, with digital signatures and report IDs noted where available), current and dated (a clear reporting period or publication date is present), specific in scope (the systems, assets, legal entities, and time windows are unambiguously defined), and relevant to one or more sub-units defined in Section 3.1 (custody and key management; change/incident management; compliance tooling; external assurance). Items are excluded if they are purely promotional with unverifiable claims, anonymous blogs or forum posts without primary-source links, expired or unverifiable certificates, or general media articles that cannot be traced back to a primary artefact.

Sourcing proceeds from primary repositories to triangulating references. Primary materials are drawn from investor-relations pages; security, legal, transparency and trust-centre hubs; status/incident portals; technical white papers; regulator registers and enforcement records; and official repositories (for example, GitHub) when platforms open-source PoR verifiers or custody specifications. Secondary sources include reputable third-party assessments (e.g., CER.live security ratings, BCG/CryptoCompare benchmarks), recognised standards and frameworks (e.g., ISO/IEC 27001, NIST SP 800-207, COBIT 2019), and supervisory guidance (e.g., ESMA/MiCA, FATF). These secondary materials are used to corroborate or challenge platform claims rather than as “ground truth.” Because web content is volatile, pages likely to change are captured as PDF/A and WARC files and archived using the Internet Archive’s “Save Page Now” or Webrecorder. For integrity, each file is accompanied by a SHA-256 hash and a

trusted timestamp (RFC-3161 or equivalent). Hash lists and capture logs are preserved in the project archive.

Where feasible, authenticity is independently verified. TLS certificates are checked via Certificate Transparency logs, the registration and standing of audit firms are confirmed through professional bodies, and PoR address sets are verified on-chain against disclosures, with clear caveats about snapshot timing and coverage. Any mismatch is coded explicitly as a limitation rather than removed silently.

Every item is catalogued in a case evidence matrix containing the source URL, as-of date, scope, systems covered, legal entities, document type, and sub-unit mapping. To enable like-for-like comparison, a short scoring rubric (0–3 per signal) is applied to each item and then rolled up at the platform level. The rubric records what is evidenced on six recurring trust signals: (i) the type and period of assurance (for example, SOC 2 Type II, ISO/IEC 27001, ISAE 3000/3402, including coverage window and system boundary); (ii) the PoR/PoL method and coverage (assets-only versus assets plus liabilities, use of Merkle-tree and/or on-chain commitments, sampling approach, and treatment of derivatives); (iii) the identity and independence of any attester (a named third party, statements of independence, and method notes); (iv) the cadence of updates (the frequency and regularity of reserve attestations, audit cycles, and status-page updates); (v) operational transparency (publication of uptime, MTTR/MTBF, time-to-post-mortem, and remediation service-level targets); and (vi) custody-control disclosures (use of HSM/MPC, vendor attestations, stated cold-storage targets, defaults for withdrawal allow-lists and 2FA, and bug-bounty scope). The score is not a normative judgment; it standardises what is actually evidenced so that signals can be aggregated into a composite “documentary trust signals” indicator for cross-case synthesis (Chapter 4) and later linked to user perceptions and behaviours (Chapter 5). Alongside this rubric, the matrix extracts quantitative fields wherever available—such as declared uptime, incident counts, typical RTO/MTTR, cold-storage percentages, bounty caps, and sanctions-screening coverage—to support normalised comparisons over the same time window.

Documents are then imported into NVivo for qualitative analysis. Coding follows a hybrid approach: a deductive frame aligned to the sub-units and to core trust constructs (structural assurances and situational normality), complemented by inductive open coding for emergent practices (for example, “real-time asset–liability ratios” or a “remediation clock”). A versioned codebook with worked examples is maintained, and coding memos document decisions over time. Reliability is managed through intra-coder stability checks (re-coding a sample after a

two-week interval) and a peer debrief on 10–15% of items to surface blind spots; disagreements are resolved against documented rules.

Language, ethics, and terms of use are handled conservatively. Non-English documents (e.g., German or French) are included; professional translation with back-translation is used for analytic excerpts, and originals are retained with the source language flagged. Proprietary items obtained under NDA are included only where permitted, redacted for sensitive details, and reported in aggregate without identifying marks. Public materials are collected in line with site terms of use; authentication is not bypassed and robots.txt is respected. All files are stored on EU-hosted, encrypted infrastructure with role-based access, in accordance with the data-management plan.

Bias is addressed explicitly in both construction and reporting. Third-party ratings are documented with their methodology and date and treated as triangulation inputs rather than definitive rankings. Where platform marketing claims conflict with audits or regulator records, the analysis defaults to the most conservative, better-substantiated source and records the discrepancy. To aid transparency and replication, the appendix will include a PRISMA-style flow (items identified, screened, excluded with reasons, included), the scoring rubric, the evidence-matrix schema, and the full hash manifest of archived artefacts.

These procedures convert a fast-moving, heterogeneous set of web artefacts into a coherent, time-stamped, and auditable evidence base. They also make the bridge to the quantitative strand explicit: the same governance signals documented here—such as the presence of PoL alongside PoR, the cadence of status reporting, and the independence of attestors—are the signals the survey later asks users about. In combination, the two strands allow the thesis to test where platform signals align—or fail to align—with user-level trust and behaviour.

3.4. Data-Collection Procedures

Data collection proceeds on three coordinated tracks so that each can challenge and contextualise the others: a documentary case corpus for each focal exchange, a cross-sectional user survey in the EEA, and a set of independent external benchmarks. All three tracks follow the same auditable loop—finding, capturing, verifying, filing, and analysing—with time-stamps, versioning, and change logs preserved end-to-end.

For the documentary cases (Coinbase, Binance, Bitstamp), the procedure begins by identifying eligible sources on official investor-relations, security, legal, transparency, and trust-centre hubs, as well as status/incident portals, technical white papers, supervisor registers and

enforcement records, and, where relevant, official repositories (e.g., GitHub) that host PoR verifiers or governance schemas. Target pages and files are captured in PDF/A and WARC formats with explicit as-of dates recorded; particularly volatile pages—such as status dashboards and reserve proofs—are also archived via the Internet Archive’s “Save Page Now” or Webrecorder to guard against link rot. Each artefact is hashed with SHA-256 and time-stamped using RFC-3161 or an equivalent mechanism, then catalogued in the exchange’s evidence matrix with its URL, capture time, scope, legal entities, systems covered, and sub-unit mapping. Materials are imported into NVivo and coded against a hybrid codebook that combines deductive nodes (aligned to the predefined sub-units and trust constructs) with inductive nodes for emergent practices. If a platform amends or replaces an audit, reserve proof, or post-mortem, both versions are retained and explicitly cross-referenced in the case narrative to preserve change history. Where feasible, PoR address sets are verified on-chain against the disclosure, auditor registrations are checked with professional bodies, and any mismatch is recorded as a limitation rather than removed.

The survey is implemented in Qualtrics (or an equivalent EU-hosted platform) and fielded between 1 February and 31 March 2025—the same window used to snapshot external benchmarks and most documentary updates. The instrument flows in a simple sequence: informed consent, screening, core constructs, platform-specific modules, and debrief. Recruitment links carry unique tokens to discourage duplication; the recruitment mode (panel versus community) is stored to enable tests for mode effects. Lists of named platforms are randomised and always include “Other (specify).” Potentially unfamiliar terms (e.g., proof-of-reserves, withdrawal allow-list) are supported with concise, in-survey tooltips to reduce comprehension bias. Two attention checks and one instructed-response item are embedded, and, where this does not harm readability, item order is randomised within constructs. The survey is offered in English and translated languages using a back-translation procedure; the response language is recorded to support measurement-invariance tests.

To anchor the link between respondents and platforms, each participant identifies a primary exchange, defined as the venue with the largest share of their traded volume in the past 90 days. A brief allocation task then records percentage shares across all used platforms, constrained to sum to 100 per cent. Multihoming is not excluded: the primary platform anchors the linkage to the documentary cases, while additional platforms later enter the models as controls.

Data hygiene follows a pre-registered protocol. Screen-fails, straight-liners or long-string responders, speeders (completion time below one-third of the median), and duplicate device fingerprints (where respondents consent to anti-fraud checks) are removed. IP addresses are not

stored. To deter automation, reCAPTCHA and a hidden honeypot field are enabled. After cleaning, the sample is compared by recruitment mode (panel versus community) and by response timing (early versus late) as simple checks for mode and non-response bias. Where material differences appear, both raw and minimally post-stratified estimates (country $\times$ age, with gender where needed) are reported as sensitivity analyses; weighted results do not replace the primary unweighted analyses without explicit justification.

Common-method and measurement safeguards are applied throughout. Beyond randomisation and neutral wording, the instrument includes a marker variable with low theoretical relatedness to the targets to gauge common-method variance. Harman’s one-factor test is reported as a descriptive screen, and a CFA/PLS-CFA with a common latent factor is estimated in sensitivity checks. For translations, configural and metric invariance are tested; if metric invariance does not hold, multi-group analyses are presented by language rather than pooled.

External benchmarks are maintained on an aligned event calendar that records governance developments such as new PoR/PoL releases, audit publications, licence grants or withdrawals, material outages, and incident post-mortems. Independent datasets—including CER.live security ratings, Chainalysis illicit-flow indicators, Coincub jurisdictional risk scores, Cornerstone enforcement tallies, and relevant ESMA/MiCA notices—are snapshotted in the same weeks as the survey and documentary captures. These sources serve for triangulation and context. Where claims conflict with primary artefacts, the analysis defaults to the most conservative, better-substantiated evidence and records the discrepancy. To control for incident salience in the models, the survey includes a short recency prompt that asks whether respondents noticed a relevant outage, reserve update, or regulatory action at their primary platform in the previous 14 days.

Integration is handled through disciplined versioning. As documents are coded, key governance signals—such as the presence of PoL alongside PoR, the identity and independence of the attester, custody disclosures (HSM/MPC), publication of uptime/MTTR, remediation service levels, and withdrawal allow-list defaults—are extracted into the evidence matrix and scored using the rubric described in Section 3.3. These signals are rolled up into a platform-level “documentary trust signals” index and aligned with each respondent’s declared primary platform for cross-strand analyses. “Other (specify)” entries are canonicalised where possible; where primary platforms fall outside the three cases, those responses remain in survey analyses but are excluded from case-linked tests and used as a comparison group. All integration scripts and change logs are version-controlled in a private Git repository to ensure full reproducibility.

Data security and ethics mirror the commitments made to respondents. Public document captures and survey data are stored on GDPR-compliant, EU-hosted infrastructure with role-based access, encryption in transit and at rest, and encrypted backups. The legal basis for the survey is consent under GDPR Article 6(1)(a). The participant information sheet provides Faculty DPO contact details and clear instructions for exercising data-subject rights, including access, deletion, and withdrawal of consent. Any optional contact information for incentives is collected in a separate, unlinkable store and deleted after fulfilment. A data-management plan specifies a retention period of 24 months after the defence, followed by cryptographic erasure. Collection respects site terms of use; authentication is not bypassed and robots.txt is honoured. Documents obtained under NDA are used only where permitted, are redacted for sensitive details, and are reported only in aggregate.

An explicit audit trail underpins transparency. The appendix will include a PRISMA-style flow diagram for the document corpus (identified → screened → excluded with reasons → included), the full hash manifest for archived artefacts, the documentary scoring rubric, the NVivo codebook, and the complete questionnaire (English and translations), together with the pre-registered cleaning and analysis plan (Open Science Framework; OSF ID to be added upon registration). Taken together, this package enables another researcher to replay the pipeline end-to-end and to verify that platform signals, user perceptions, and independent benchmarks were aligned on the same timeline and merged in a way that makes alignment—and misalignment—between signals and behaviour empirically testable.

3.5. Analytical Techniques

The study follows a convergent–parallel strategy: the documentary cases and the survey are analysed separately with methods suited to each strand and then brought together in a structured integration step. The objective is not to force the strands to agree but to assess alignment—whether governance signals published by platforms correspond to what users perceive and how they behave—and to treat any misalignment as a substantive result rather than an error (Creswell & Plano Clark, 2018).

On the qualitative side, the documentary corpus is examined using reflexive thematic analysis in the sense of Braun and Clarke (2022). The analysis proceeds through familiarisation, systematic coding, theme development, review, and write-up. Coding begins from a deductive scaffold grounded in trust theory: institution-based trust (structural assurances such as audits,

licences, and reserve proofs; and situational normality such as operational stability and predictable user experience), trusting beliefs (competence, integrity, benevolence, and communication), and artifact-specific signals salient in FinTech (for example, the design of proof-of-reserves and proof-of-liabilities, the identity and independence of the attester, publication of uptime/SLO histories, the depth and timeliness of incident post-mortems, custody architecture such as HSM/MPC and key-ceremony practice, and change/vendor-risk controls). The coding frame remains open to inductive themes that often surface only in technical artefacts—timelock defaults, oracle diversity, circuit-breaker constraints, or remediation service levels. Within each case, pattern matching and explanation building test hypothesised mechanisms (e.g., PoR + PoL with an independent attester increasing the credibility of solvency signals) against the record (Yin, 2018). Cross-case synthesis uses structured matrices so that convergence and divergence are visible—how precisely custody is documented, how candid post-mortems are, or how vendor risk is governed.

To aid comparability, a compact documentary trust-signals index is constructed. Item-level extractions—such as the presence and quality of PoR+PoL, the checkability of verifier identity and independence, publication of uptime/MTTR with history, evidence of incident remediation, defaults for withdrawal allow-lists, and related cues—are summed and rescaled to 0–100. Because these indicators *form* the composite rather than reflect a single latent factor, the index is treated as a formative construct and is not subjected to internal-consistency tests. Content validity is supported by a rubric published in the appendix; all binary extractions that feed the index undergo a second-pass verification by a methods advisor, with disagreements resolved by tightening the rubric and documenting the decision. An auditable trail—codebook, change log, and hash manifest—underpins transparency.

The quantitative strand advances in five stages. First, screening and missingness diagnostics are performed; within multi-item constructs, person-mean imputation is applied when at most one item is missing, otherwise listwise deletion is used, with multiple-imputation sensitivity checks on key models. Highly skewed behavioural variables (e.g., trade counts) are log-transformed in robustness checks. Second, measurement is validated. Where items were adapted to the crypto context, an exploratory factor analysis (EFA) on a random half-sample is followed by confirmatory factor analysis (CFA) on the hold-out. Internal consistency is assessed via Cronbach's alpha and composite reliability; convergent validity via AVE ($\geq .50$), and discriminant validity via HTMT ($< .85$) (Hair et al., 2021). For multilingual deployment, configural and metric invariance are tested; if metric invariance does not hold, results are presented by language group rather than pooled. Third, descriptive profiles of trust, risk,

UTAUT2 antecedents, and behaviours are produced; platform-level profiles average respondents' scores for their declared primary platform and display z-standardised composites alongside tables.

Fourth, associations and predictions are estimated with appropriate attention to nesting. Bivariate associations (Pearson/Spearman) map the terrain. Multivariable models then estimate the extent to which visible safeguards and institutional assurances are associated with trusting beliefs, intentions, and behaviour, controlling for UTAUT2 factors (performance and effort expectancy, social influence, habit, facilitating conditions), demographics, experience, risk tolerance, digital literacy, and multihoming. Because respondents are nested within platforms, models use multilevel (mixed-effects) regression with a random intercept for primary platform—and, in sensitivity checks, for country—or at minimum cluster-robust standard errors by platform. Where theory proposes a path—structural assurances → trusting beliefs → intention → behaviour—a PLS-SEM is estimated to accommodate complex models and common distributional quirks in survey data. Latent constructs are modelled reflectively; the documentary index enters as an observed exogenous formative composite. Model quality is reported with R^2 , Stone–Geisser Q^2 , SRMR/NFI, bootstrapped indirect effects for mediation, and VIF (< 3.3). Effect sizes (standardised β , f^2 for SEM, partial η^2 for group tests) and 95% confidence intervals are reported; families of related hypotheses are controlled using Benjamini–Hochberg.

Fifth, heterogeneity and event sensitivity are examined. Moderation and subgroup analyses explore differences by age, gender, experience, country group, and language. Interaction terms test, for example, whether the association between documentary assurances and trusting beliefs is stronger for less-experienced traders. An event-sensitivity probe leverages the aligned calendar: respondents who report noticing a recent governance event (e.g., a new audit or reserve update) are contrasted with those who did not on nearby trust items, optionally adjusting for recency, demographics, and baseline risk tolerance. Given the cross-sectional design, these contrasts are exploratory and interpreted cautiously.

Integration across strands uses joint displays that link platform-level documentary signals (the 0–100 index) and external benchmarks to user-level perceptions and behaviours. Two complementary views are produced: ecological correlations between the documentary index and platform-aggregated user trust/behaviour (with bootstrapped confidence intervals), and individual-level mixed-effects regressions in which each respondent is linked to the index of their primary platform. Where strands align (“confirmation”), the case narratives trace plausible mechanisms; where they complement one another (“complementarity”), the display shows how

the pieces fit; and where they diverge (“discordance”)—for instance, strong documentary signals that users do not notice in the interface—the gap is treated as a design-relevant finding. All integration decisions (weighting choices, clustering, alternative index weightings) are logged and reported as sensitivity checks.

Reproducibility is ensured through preregistration and full materials. Code for cleaning, scoring, modelling, and joint displays is version-controlled in a private Git repository and preregistered on the Open Science Framework. The appendix publishes the full questionnaire, translation notes, the codebook, the scoring rubric, and a hash-verified manifest of all documentary artefacts. Together, these materials allow another researcher to replay the pipeline end-to-end and to verify that platform signals, user perceptions, and independent benchmarks were aligned on the same timeline and merged in a way that makes alignment—and misalignment—between signals and behaviour empirically testable.

3.6. Limitations and Mitigation Measures

This design makes deliberate trade-offs to balance realism, feasibility, and rigour. The most visible choice is the exclusion of interviews. That inevitably forgoes some insider nuance. To offset this loss, the documentary cases are widened to include artefacts that often speak more plainly than press remarks—formal incident post-mortems, security audit letters, reserve-proof schemas, change-control runbooks, and, where lawfully available, redacted risk-committee materials. In parallel, independent benchmarks (for example, security ratings and enforcement tallies) are aligned to the same timeline to temper platform self-portrayals. The consequence is less narrative colour but more auditable evidence.

A second limitation concerns document bias, opacity, and survivorship. Public artefacts can be selective or promotional, and failed or opaque platforms tend to leave a thinner paper trail. Mitigation rests on strict inclusion criteria (authentic, date-stamped, scope-clear, and relevant), source triangulation (official sites, regulator registers, reputable third parties), and a content hierarchy that privileges verifiable evidence—such as audit scope, attestor identity and independence, liabilities coverage, and uptime histories—over marketing claims. To reduce survivorship bias, the corpus incorporates archival captures (for example, the Internet Archive) and regulator filings for delisted or defunct entities where available; nonetheless, the bias cannot be eliminated entirely. All captured items are hashed and versioned so that mid-study changes (for instance, an amended proof or updated audit letter) remain traceable.

The survey strand inherits familiar risks of self-report and common-method variance. Perceptions are noisy, and measuring predictors and outcomes in a single instrument can inflate associations. Several precautions are used: neutral wording, mixed item direction, two attention checks plus an instructed-response item, and a questionnaire flow that temporally separates key predictors from behavioural outcomes. In analysis, method artefacts are probed diagnostically (variance-inflation factors, a marker variable, and a Harman single-factor screen). Behavioural questions are anchored to recent windows (30/90 days) to reduce recall error, and extreme values are examined with and without winsorisation. Because respondents may apply different “personal scales” when rating platforms, descriptive platform profiles are standardised to z-scores and interpreted as relative rather than absolute differences.

Sampling bias is another concern. Online recruitment can over-represent heavier, more engaged traders. Country quotas across the EEA, screening for recent activity (≥ 1 trade in the last 90 days), and optional post-stratification weights (country $\times$ age, and gender where needed) help to correct imbalances; key models are re-estimated by experience tier and multihoming status as robustness checks. To curb panel fraud and duplication, recruitment links use one-time tokens and IP throttling, and—where permitted and consented—light-touch device fingerprinting; suspicious records are flagged automatically and reviewed manually. Because respondents self-declare their primary platform, a short consistency check (for example, last transaction or product type) validates that identification; inconsistent cases are retained only for sensitivity analyses.

The cross-sectional design limits causal inference. Associations are interpreted through theory and triangulation, not as proofs of effect. Where fieldwork overlaps with governance events (for example, a new audit or a major outage), exploratory contrasts between respondents who noticed the event and those who did not are reported and clearly labelled as non-causal.

Language and translation can introduce variation. To minimise drift, the instrument undergoes translation/back-translation with bilingual spot-checks; in analysis, measurement invariance is tested across languages. If metric invariance does not hold, results are reported by language rather than pooled.

Respondents are nested within platforms and select those platforms non-randomly. Primary models therefore use multilevel (mixed-effects) regression with random intercepts for platform (and, in sensitivity checks, for country), or at minimum cluster-robust standard errors by platform. Ecological correlations between the documentary index and platform-aggregated user trust are presented as descriptive context only, given the risk of ecological fallacy; confirmatory

tests occur at the individual level with appropriate clustering and controls (experience, risk tolerance, digital literacy, and multihoming).

Two modelling cautions apply. First, the documentary trust-signals index is a formative composite: items *cause* the score rather than reflect a single latent construct. It is therefore not subjected to internal-consistency tests. Subjectivity is reduced by publishing the scoring rubric, double-checking binary extractions, and running leave-one-indicator-out checks alongside alternative weighting schemes (equal weights versus frequency-of-evidence weights). Second, to avoid overfitting in regressions and PLS-SEM, models remain parsimonious as pre-registered; where more complex paths are explored, they are checked on a hold-out subsample and families of related hypotheses are controlled using Benjamini–Hochberg procedures.

Rapid market change introduces timing risk. A reserve-proof schema, status page, or licence can change mid-study. A governance-event calendar and a versioned evidence archive keep analyses time-aware; each claim in the case write-ups carries an explicit as-of date. When artefacts are replaced, both versions are retained and coded so the change history remains visible.

Finally, data protection and access set hard boundaries. GDPR compliance rules out direct identifiers; proprietary documents are used only where access is lawful and ethical, and any sensitive passages are anonymised. Some high-value artefacts (for example, full SOC workpapers) are unavailable by design; in such cases the study relies on regulator filings or independent summaries and flags the limitation.

Even with these measures, some biases—such as survivorship in documents or self-selection in users—remain possible. For that reason, all substantive findings are presented with sensitivity analyses, confidence intervals, and clear as-of dates, enabling readers to judge how far, and under what conditions, the conclusions can travel.

4. RESULTS

4.1. Platform Profiles and Governance Structures

This section profiles three exchanges—Coinbase, Binance, and Bitstamp—to illustrate how different governance choices are turned into user-visible assurances. Read through the lens of trust theory, the cases show how structural assurances and “situational normality” cues can reinforce both institution-based trust and trust in technology (McKnight & Chervany, 2001).

Coinbase. Coinbase operates as a publicly listed U.S. company and pairs regulatory authorisation with independent control attestations and unusually visible operations. Institutionally, Coinbase Custody Trust Company has been authorised by the New York State Department of Financial Services (NYDFS) since 23 October 2018 to act as a New York limited-purpose trust company, enabling qualified-custodian activities (New York State Department of Financial Services, 2018). Control assurance includes SOC 1 Type II and SOC 2 Type II examinations for institutional units (e.g., Prime/Custody) and, more recently, a 2025 SOC 2 Type 1 for staking, all complemented by published security practices (Coinbase, 2022, 2025a). Operationally, a live status page with an incident archive functions as a standing cue of reliability. In May 2025, Coinbase publicly disclosed an extortion/social-engineering incident targeting support contractors, outlined mitigations and reimbursement policies, and documented the event in its incident communications—an example of candid post-incident transparency (Coinbase Global, Inc., 2025; Coinbase, 2025b). Taken together, these artefacts signal strong institution-based trust (licensing, third-party attestations) and trust in the firm’s technology stack (clear reliability and security controls) (McKnight & Chervany, 2001).

Binance. Binance’s governance posture was reshaped by a November 2023 U.S. criminal resolution: the company and its founder pleaded guilty to violations related to the Bank Secrecy Act and sanctions, the founder stepped down as CEO, and the settlement—approximately \$4.3 billion—installed an independent compliance monitor (U.S. Department of Justice, 2023). In

the period since, corporate communications have emphasised compliance upgrades and greater operational transparency. On solvency signalling, Binance operates a Proof-of-Reserves (PoR) portal based on Merkle-tree attestations and provides educational explainers of the method (Binance, 2022). While such disclosures can lift perceived transparency, PoR is a point-in-time view of assets and typically does not capture off-chain liabilities; it is therefore not equivalent to a full audit. In trust terms, the mix of enforcement-driven remediation, status transparency, and cryptographic proofs aims to rebuild institution-based trust under oversight while offering technology-level signals, albeit with clear scope limits.

Bitstamp. Bitstamp, one of the longest-operating exchanges (founded 2011), closed its acquisition by Robinhood on 2 June 2025, bringing the firm under a U.S. public-company umbrella (Robinhood Markets, Inc., 2025). Its transparency posture includes regular status and maintenance communications and governance updates aimed at retail users. Structural assurances feature ISO/IEC 27001 certification and SOC 2 Type II reports, alongside messaging that annual financial statements are audited by a Big Four firm. Notably, Bitstamp also publishes plain-language explainers that distinguish PoR from broader “proof-of-solvency” concepts that incorporate liabilities—an effort to make scope and limitations clear to non-experts (Bitstamp, 2022). This combination of third-party assurance and accessible communication strengthens both structural and communication-driven antecedents of trust (McKnight & Chervany, 2001).

Across the three cases, different paths lead to similar trust mechanisms. Coinbase leans on licensing, recurring attestations, and practiced disclosure; Binance couples remediation under a monitor with cryptographic proofs; Bitstamp emphasises control certifications and explanation. The contrasts underscore two practical points for the thesis: first, that user-visible artefacts (licences, SOC/ISO reports, status histories, PoR/PoL scope notes) are the currency of institution-based trust in crypto markets; and second, that legibility—plain statements of what a proof covers and what it does not—is as important as the underlying control for sustaining trust in technology-mediated services (McKnight & Chervany, 2001).

4.2. Survey Descriptive Statistics and Correlations

The analytic dataset comprises $N = 202$ valid respondents. Items use 1–5 Likert scales (1 = strongly disagree; 5 = strongly agree). Missing data were handled via pairwise deletion.

Analytic choices follow recommended practice for survey scale reliability and correlational analysis (Hair et al., 2021).

Table 1. Sample composition

Category	Group	n
Age	18–24	87
	25–34	61
	35–44	16
	45+	38
Gender	Female	113
	Male	86
	Prefer not to say	1
Crypto experience	Never used	115
	< 1 year	37
	1–3 years	24
	> 3 years	26

Source: Author's analysis based on the study's findings

The sample skews younger (18–34) and includes a large novice group alongside experienced traders.

Table 2. Scale reliability and descriptives (1–5)

Scale	k	α	N	Mean	SD	P25	Median	P75
Knowledge self-efficacy	2	.886	202	2.81	1.39	1.50	2.50	4.00
Platform trust & security	4	.916	201	2.99	1.21	2.00	3.00	4.00
Risk perception	4	.830	202	3.15	0.95	2.50	3.00	3.75

Switching behaviour	4	.806	201	3.84	0.88	3.25	4.00	4.50
Transparency expectations	2	.672	201	3.96	1.04	3.00	4.00	5.00

Source: Author's analysis based on the study's findings

Transparency expectations and switching readiness are high (≈ 3.8 – 4.0). Knowledge and platform trust are moderate (≈ 2.8 – 3.0).

Table 3. Headline item statistics (selected; 1–5 Likert)

Item (abridged)	M	SD	Agree %	n
Trust: platform is technically secure	2.91	1.39	35.0	200
Trust: protects personal data	3.00	1.38	37.5	200
Trust: protects funds	2.97	1.44	37.2	199
Trust: transparent about data	2.96	1.39	34.5	200
Important: clear security communication	3.95	1.17	62.0	200
Platforms should educate users more	3.86	1.22	58.5	200
Concern: data misuse	3.07	1.41	43.5	200
Concern: regulatory change	3.09	1.41	43.0	200
Self-efficacy: enough knowledge	2.66	1.44	36.1	202

Source: Author's analysis based on the study's findings

Baseline trust in security/data protection is mid-range and polarized; expectations for transparent communication and education are high.

Composite index

A four-item Platform Trust composite (technical security, data protection, funds protection, data transparency) averages available items with a ≥ 3 -of-4 rule. $\alpha = .916$; $M = 2.95$, $SD = 1.27$, Median = 3.00 ($n = 198$), supporting treatment as a coherent trust construct (McKnight & Chervany, 2001; Hair et al., 2021).

Associations among constructs

Trust beliefs co-move and track self-efficacy strongly: *Technical security* with *personal-data trust* $r = .551$; with *funds-protection* $r = .563$. *Self-efficacy* with *technical security* $r = .731$, *personal-data trust* $r = .628$, *funds protection* $r = .491$ (all $p < .001$).

Abstract risks are weakly linked to platform trust: *data-misuse risk* $\approx -.01$ to $-.03$ (n.s.); *regulatory risk* $\approx .07-.09$ (typically n.s.).

Table 4. Spearman correlations with the outcome

Outcome: “*My trust in the platform’s ICT management influences which platform I choose.*”

Predictor scale	ρ
Communication views	0.552
Switching behaviour	0.432
Transparency expectations	0.371
Knowledge self-efficacy	0.307
Platform trust & security	0.271
Risk perception	0.193

Source: Author’s analysis based on the study’s findings

Table 5. Exploratory multivariate model (standardized OLS)

Outcome: *trust influences platform choice.*

Predictor	β	p
Communication views	0.497	<.001
Switching behaviour	0.178	.051
Transparency expectations	0.075	.342
Knowledge self-efficacy	0.011	.891
Platform trust & security	−0.084	.289

Risk perception	-0.014	.822
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Source: Author's analysis based on the study's findings

Communication (feeling informed/educated) is the strongest unique correlate of trust-driven platform choice ($\beta \approx .50$).

The survey reveals a trust gap (midpoint means; polarization) alongside strong demand for clear security communication and user education. Self-efficacy is tightly associated with platform trust, echoing competence-based antecedents in technology-use models (McKnight & Chervany, 2001; Venkatesh et al., 2012).

4.3. Documentary Themes and Illustrative Evidence

The qualitative strand is a documentary analysis of publicly available and (where permitted) proprietary artefacts: certifications, proof-of-reserves/proof-of-liabilities materials, incident post-mortems, compliance disclosures, status-page archives, and policy documents. All materials were coded in NVivo following Braun and Clarke's reflexive thematic procedure (2022).

Table 6. Theme overview and typical evidence

Theme	Working definition	Typical artifacts	Links to RQs
Security architecture & custody	Design and operation of technical safeguards for asset custody/system protection	HSM & key-management docs; cold-storage thresholds; multisig; bug-bounty/red-team reports; SOC/ISO certificates	RQ1–RQ2
Privacy & data governance	Lawful, minimal, transparent handling of personal data	DPIAs; retention schedules; encryption/DLP; change-logs	RQ1, RQ3

Transparency & incident communication	Communication of risk posture, incidents, and reserves/liabilities	Status pages; incident post-mortems; PoR with liabilities; audit attestations	RQ1–RQ3
Compliance & monitoring	Tooling and governance for regulatory adherence/FC controls	AML/KYC/KYB systems; sanctions screening; travel-rule tooling; MiCA readiness	RQ1, RQ3, RQ5
User education & safety communication	How platforms teach and warn users	Risk hubs; phishing alerts; security UX patterns; change notifications	RQ3–RQ5

Source: Author’s analysis based on the study’s findings

Across the corpus, several documents made concrete, checkable commitments. One custody policy states that at least ninety-five percent of client digital assets are held in offline, multisignature storage with quarterly key rotation (Case corpus: Custody Policy, as of 2025-02-12). An incident-management statement commits to disclosing all security incidents within seventy-two hours, including a dated timeline, root cause, and remediation milestones (Case corpus: Status page / Incident post-mortem, as of 2025-03-04). A privacy update records a reduction of the data-retention window from twenty-four to twelve months and the introduction of differential access logging (Case corpus: Privacy Policy / DPIA summary, as of 2025-01-20). Taken together, these artefacts translate back-office governance into user-visible signals of competence, integrity, and reliable communication, aligning with the survey finding that transparency and clear explanations are central to trust-driven platform choice (see §4.2; McKnight and Chervany, 2001).

4.4. Cross-Case Synthesis

Segments are treated as analytic “cases” to identify convergent/divergent patterns in perceived governance and trust. Non-parametric tests (Kruskal–Wallis) complement means.

Table 7. Means by crypto experience (1–5)

Experience	Knowledge	Trust & security	Risk	Switching	Transparency
Never used	2.00	2.39	3.23	3.66	3.87

< 1 year	3.43	3.34	3.04	3.88	3.82
1–3 years	4.04	3.84	3.21	4.16	4.00
> 3 years	4.38	4.33	2.91	4.33	4.52

Source: Author's analysis based on the study's findings

Table 8. Experience gradient and tests

Scale	Δ (most experienced – novice)	H	p	Sig.
Knowledge self-efficacy	+2.38	96.450	<.001	✓
Platform trust & security	+1.94	77.771	<.001	✓
Risk perception	−0.32	2.410	.492	—
Switching behaviour	+0.67	17.145	<.001	✓
Transparency expectations	+0.65	8.846	.031	✓

Source: Author's analysis based on the study's findings

With experience, knowledge and trust rise sharply; risk perception remains broadly stable.

Table 9. Means by gender (1–5)

Gender	Knowledge	Trust & security	Risk	Switching	Transparency
Female	2.40	2.82	3.29	3.85	4.07
Male	3.35	3.21	2.96	3.81	3.80
Prefer not to say	2.50	1.75	2.50	4.25	5.00

Source: Author's analysis based on the study's findings

Table 10. Gender tests (Kruskal–Wallis)

Scale	H	p	Significant?
Knowledge self-efficacy	22.501	<.001	Yes
Platform trust & security	6.732	.035	Yes

Risk perception	7.320	.026	Yes
Switching behaviour	0.337	.845	No
Transparency expectations	3.627	.163	No

Source: Author's analysis based on the study's findings

Table 11. Age tests (Kruskal–Wallis)

Scale	H	p	Significant?
Knowledge self-efficacy	7.050	.070	No (trend)
Platform trust & security	3.558	.313	No
Risk perception	1.151	.765	No
Switching behaviour	3.899	.273	No
Transparency expectations	0.865	.834	No

Source: Author's analysis based on the study's findings

Across respondent segments, expectations for transparency are uniformly high and rise modestly with experience. Readiness to switch providers is also high and increases with greater trading experience. Perceptions of platform trust and security sit at a moderate level overall but climb sharply along the experience gradient, and women report somewhat lower trust than men. Self-assessed knowledge follows the same pattern: confidence grows markedly with experience, while women, on average, report lower self-efficacy. By contrast, perceived risk remains close to the midpoint and shows little variation across groups (see Exhibits 4.4A to 4.4E).

Taken together, these patterns suggest a straightforward managerial lever. Robust security and custody practices should be paired with plain, frequent communication and targeted education, especially for novices and other under-confident cohorts, to convert governance quality into trust-based platform selection and sustained use. In practice, this means maintaining visible status and incident updates, explaining solvency and custody in accessible language, and offering inclusive onboarding that builds confidence without diluting technical substance.

4.5. Triangulation of Quantitative and Qualitative Findings

Table 12. Construct-level triangulation (survey ↔ documentary/secondary)

Lever	Survey evidence	Documentary/secondary evidence (status)	Meta-inference
Transparency & communication	High expectations (Mean $\approx$ 3.96); strongest correlate of choice ($\rho = .552$; $\beta \approx .50$)	Presence of status pages, incident post-mortems, PoR explainers varies by platform	Convergent when artifacts are visible; transparency is a primary trust lever
Security & custody	Trust moderate (M $\approx$ 2.99); 73.5% would switch on poor security reputation	SOC/ISO certificates, custody whitepapers, bounty reports	Complementary: strong controls need plain communication to raise perceived trust
Privacy & data governance	70.6% willing to adopt better-privacy platforms; personal-data trust mid-range	Retention schedules, DPIAs, change-logs vary in legibility	Convergent if specific and user-readable; otherwise perception gap persists
Compliance & monitoring	Regulatory risk perceived (Mean $\approx$ 3.06)	Enforcement resolutions/monitoring; travel-rule tooling disclosures	Visible compliance reduces uncertainty; impact mediated by clarity
User education	68.2% want more ICT-risk education; communication predicts choice	Risk hubs, alerts, onboarding checklists	Education connects governance to perceived safety and platform choice

Source: Author's analysis based on the study's findings

The robustness and validity checks indicate that four multi-item scales achieved good internal consistency ($\alpha \approx .81-.92$), while the two-item transparency index was acceptable for a short scale ($\alpha = .672$). By contrast, the brief communication mini-scale showed low reliability ($\alpha = .492$) and is interpreted with caution. Given the ordinal nature of the items, the analysis reports Spearman's ρ ; Pearson's r produced the same ordering of effects. Group comparisons reveal pronounced experience gradients for both knowledge and trust, as well as gender differences in knowledge, trust, and risk perception; no statistically reliable age effects were detected. In multivariate models, perceived communication quality remained the strongest unique correlate of whether trust influenced platform choice ($\beta \approx .50, p < .001$), with switching propensity showing a weaker, marginal association ($\beta \approx .18, p \approx .051$). Item-level missingness was uneven—most notably for the incident-reporting statement ($n = 94$) and terms-and-conditions awareness ($n = 173$)—and is flagged in the notes. Common-method variance was mitigated procedurally (mixed item directions, attention checks) and probed statistically, but some residual risk remains without external validation.

Two practical implications follow for ICT management. First, governance needs to be legible as well as strong: platforms should explain security, privacy, and incident handling in plain language and keep evidence current and date-stamped so users can verify claims. Second, because security reputation is tightly linked to user mobility, preventive controls should be paired with candid incident disclosures to preserve reputation capital. Complementing these foundations, sustained investment in user education—risk hubs, proactive alerts, and concise onboarding checklists—can lift trust, particularly for under-confident cohorts. Inclusive onboarding, progressive security defaults (e.g., easy MFA), and clear summaries of key terms help reduce ambiguity for these groups. Finally, given the persistent salience of risk in this category, protective defaults such as withdrawal allow-lists and sensible rate limits should remain enabled, and critical warnings should be repeated at decision points to reinforce safe behaviour.

Across items, respondents put safety and clarity ahead of everything else. Nearly three quarters (73.5%) said they would leave a platform with a poor security reputation. A similar share want the basics explained plainly—71.6% prioritised clear communication about data protection—and 70.6% would switch to a provider that demonstrably protects data better. Many also asked for help to use platforms safely: 68.2% felt exchanges should invest more in ICT-risk education. Cost matters, but less than safety; 59.7% would move because of high fees.

Where the picture is weaker is in baseline understanding and trust. Only 27.2% felt well informed about terms and conditions, and just a third (33.2%) believed they had enough

knowledge to use platforms safely. Fewer than four in ten expressed confidence that their platform is technically secure (36.8%) or protects personal data (37.3%), while 35.5% acknowledged a meaningful risk of data misuse. Put simply, users are ready to reward credible security and straight communication, but many do not yet feel informed or confident enough to take those reassurances at face value—especially newer or less self-assured traders.

5. DISCUSSION

5.1. Interpreting Results through Trust Theory

Viewed through contemporary trust theory in FinTech and information systems, the results point to a clear pattern in what users need before trust meaningfully shapes platform choice. Respondents report strong expectations for transparency and clear communication and say they are ready to switch if those expectations are not met. At the same time, baseline trust in technical security, personal-data protection, and fund safety is only moderate. In McKnight and Chervany's terms, this gap sits between institution-based trust—the sense that credible rules, safeguards, and everyday normality are in place—and trusting beliefs about a specific platform's competence and integrity (McKnight & Chervany, 2001). In practice, users seem persuadable, but they want to see governance in action—evidence of controls and candid incident communication—before converting that openness into commitment.

A second, consistent signal is the close link between self-efficacy and platform-directed trust. Participants who feel more knowledgeable also express higher trust across security and protection dimensions. This aligns with the view that, in FinTech, the “trustee” is not only an organisation but also an IT artefact—applications, proofs, models, and protocols—whose reliability must be interpreted by users with uneven literacy (Lacity et al., 2024). Experience appears to build that interpretive capacity: the clear novice-to-expert gradient in both self-

efficacy and trust suggests that competence lowers perceived vulnerability and makes both artefact-level and organisational assurances easier to read (McKnight & Chervany, 2001).

The findings also move communication from a supporting role to a proximal driver of choice. Perceived communication quality—feeling informed and educated—shows the strongest unique association with whether trust actually affects platform selection ($\beta \approx 0.50$), even after accounting for overall trust composites and risk perceptions. This sits comfortably with trust theory, where communication feeds the formation of trusting beliefs and intentions, and with adoption models that highlight facilitating conditions and credible service signals as levers of intention and use (McKnight & Chervany, 2001; Venkatesh, Thong, & Xu, 2012; Amnas et al., 2023). In short, how platforms explain their safeguards appears to matter at least as much as the safeguards themselves.

By contrast, abstract risks—such as general privacy concerns or regulatory change—show only weak ties to artefact-level trust. Respondents seem to distinguish exogenous uncertainty (policy shifts, category-level privacy debates) from endogenous platform safety. This separation points to a two-track governance strategy: keep improving internal controls and make them legible, while acknowledging external uncertainty without over-promising control (Devlin et al., 2025; Lacity et al., 2024). The more credible commitment becomes: “Here is what the platform can guarantee; here is what no provider can fully control.”

Heterogeneity in how signals are received also matters. Gender differences are visible: on average, women report lower self-efficacy and trust and higher perceived risk than men, despite similar expectations for transparency. The literature anticipates such variation and recommends communication and onboarding that reduce cognitive load for under-confident segments without diluting technical content—progressive disclosure, default-safe settings, and short, scenario-based guides that tie controls to concrete “what if” situations (Amnas et al., 2023).

Finally, the coexistence of a trust gap and a communication premium helps reconcile survey polarisation with the documentary record. Trust in technology and institution-based trust are co-produced in FinTech. Strong back-office controls are necessary but insufficient unless they are translated into user-facing proofs—status histories with post-mortems, scope-clear attestations, and solvency evidence that covers both reserves and liabilities (Devlin et al., 2025; Lacity et al., 2024). The pattern that emerges is theoretically tidy: structural assurances and situational normality set the floor; human-understandable communication raises the ceiling; and user competence determines how high that ceiling can go for different segments (McKnight & Chervany, 2001; Venkatesh et al., 2012; Amnas et al., 2023).

5.2. Implications for ICT Governance and Management

The results point to a straightforward managerial mandate: governance must be made legible, not merely robust. Users reward safeguards they can see and understand—clear status information, scoped attestations, and plain explanations—more than abstract assurances. In trust-theoretic terms, structural assurances and situational normality only translate into platform choice when they are rendered as user-facing proofs that support trusting beliefs and intentions (McKnight & Chervany, 2001; Lacity et al., 2024; Devlin et al., 2025).

Translating controls into public evidence is the first task. Operational transparency should be routine rather than episodic: a live status page with historical uptime, time-to-detect and time-to-restore metrics, and dated post-mortems that state root cause, blast radius, and remediation milestones sets expectations for “how things work here.” Committing to disclosure service levels—an initial notice within hours and a full post-mortem within roughly a week—helps to normalise incident communication and reinforces situational normality (McKnight & Chervany, 2001). Security and custody practices should be documented with scope-clear attestations (e.g., SOC 2, ISO/IEC 27001), summaries of key-management architecture (HSM/MPC, cold-storage thresholds), change-control routines, and concise reports from bug-bounty programmes. On solvency signalling, proof-of-reserves should be paired with proof-of-liabilities and a named independent attester, with scope and cadence stated to avoid “assets-only” signalling (Lacity et al., 2024; Devlin et al., 2025). Because many users will not read long technical documents, platforms should also offer plain-language “risk nutrition labels” that distinguish market, technical/operational, and counterparty risks, clarify who bears which risks under what conditions, and point to deeper artefacts. Education should be treated as product infrastructure: short, scenario-based explainers—such as what happens after a device is lost or how withdrawals are protected—should be tied directly to controls and settings so that safeguards are not only present but actionable (Venkatesh, Thong, & Xu, 2012; Amnas et al., 2023).

Institutional anchoring makes this transparency durable. Public artefacts should be mapped to recognised frameworks—most notably NIST CSF 2.0, with emphasis on the GOVERN function that sets policy, roles, risk appetite, and supply-chain oversight, and ISO/IEC 27001:2022 control families—so that evidence is reusable, auditable, and stable across releases (NIST, 2024; ISO, 2022). For EU-exposed firms, DORA provides a ready template for incident classification and reporting, resilience testing, and oversight of critical third parties; publishing testing cadence and major-incident thresholds brings those obligations into public view

(EIOPA, 2025; ESMA, 2025). Claims about solvency and safeguarding should be explicitly linked to routes of recourse and complaints handling, so that structural assurances are paired with practical remedies.

Design also needs to reflect user heterogeneity in competence and confidence. Novices benefit from progressive disclosure, default-safe settings such as withdrawal allow-lists enabled by default and MFA prompts at first high-risk actions, and concise summaries of terms and conditions. More experienced users expect depth: access to full audit letters, method notes for PoR/PoL, on-chain verifiers, and clear scope statements—without losing the plain-language synopsis. Where confidence gaps are visible (for example, by gender in this sample), platforms can provide checklists that deliver quick safety wins—successful MFA setup, allow-list confirmation, recovery-kit rehearsal—alongside visible feedback loops (“last security check: OK”), which reduce perceived vulnerability without diluting technical content (Amnas et al., 2023).

Finally, governance should be managed—and shown—through a small set of leading indicators. Executives should review trust-oriented KPIs at the same cadence as product metrics, including time-to-post-mortem, the share of incidents with remediation milestones closed on time, attestation freshness and scope coverage, the cadence and liabilities coverage of PoR+PoL, education reach and its lift on safe-configuration completion or retention, and the share of users on default-safe controls (e.g., MFA, allow-lists). Packaging these signals in a quarterly Trust & Resilience Report turns internal discipline into durable, outward-facing evidence (McKnight & Chervany, 2001). As a practical 90-day deliverable, a versioned “governance assurance pack” can bundle the status-page history and disclosure SLA; current SOC/ISO attestations with scope notes; custody and key-management summaries; PoR+PoL with a named, independent attester and stated liabilities method; the incident-response playbook with classification thresholds and timelines; a register of critical third-party dependencies; and the risk-label screen with scenario explainers. Each element can be mapped to NIST CSF 2.0 GOVERN and ISO/IEC 27001 controls and, where relevant, to DORA obligations (NIST, 2024; ISO, 2022; EIOPA, 2025; ESMA, 2025).

In sum, the managerial path is evidence-led: convert strong back-office practice into legible front-stage proof, anchor it in recognised frameworks, tailor communication to varied user competencies, and manage by trust KPIs. These moves act along the same pathways highlighted in the results—raising perceived service quality, lowering perceived risk, and turning trust into platform choice and continuance (McKnight & Chervany, 2001; Venkatesh et al., 2012; Amnas et al., 2023).

5.3. Regulatory and Policy Implications

Taken together, the results point to a simple idea with wide policy relevance: users reward platforms that “show their homework.” They do not need slogans; they need concrete, checkable evidence that governance is real. This creates room for regulators to raise trust at relatively low cost by standardising what evidence should be public and how it should be presented.

A practical starting point is a short, standardised governance assurance pack that every centralised exchange—and any retail-facing DeFi front end—keeps current. In one, easily located place, users should see proof-of-reserves paired with proof-of-liabilities, the identity and independence of the attester, a plain explanation of methods and scope, and the cadence of updates. Beside solvency evidence, platforms should present scope-clear SOC/ISO attestations, a concise summary of custody design (e.g., HSM/MPC and cold-storage targets), a status-page archive with uptime and dated post-mortems, and an incident-response policy with explicit disclosure timelines. Publishing this pack in a lightweight, machine-readable schema would make assurances comparable across firms and borders, in line with the direction of contemporary international guidance (IOSCO, 2023; FSB, 2023a, 2023b; ISO, 2022).

Quality and cadence matter as much as presence. Rules should require clear “as-of” dates, defined system boundaries, explicit liabilities coverage, the attester’s name and independence, and a predictable update schedule. These basics curb “trust theatre” and reflect how people actually decide: few retail users will read a 100-page audit, but many will grasp a dated post-mortem with a closed-out remediation plan (Devlin et al., 2025; Lacity et al., 2024). In the European Union, supervisors can anchor these expectations in what DORA already demands—incident classification and reporting, resilience testing, and oversight of critical third parties—while encouraging firms to convert those duties into user-visible signals (EIOPA, 2025; ESMA, 2025).

Privacy should be treated the same way: not as a promise, but as artefacts. Short, published retention schedules, change-logs for policy updates, and plain-language DPIA summaries make lawful basis and minimisation tangible. Because this study shows that communication and education convert trust into platform choice, policy should support plain-language risk labels for common protocol classes and custody models, test them for comprehension, and allow proportionate “comply or explain” adoption so designs can improve without freezing innovation (Amnas et al., 2023; IOSCO, 2023).

Given the centrality of solvency signalling, guidance should also clarify what “good” looks like for PoR/PoL: acceptable assurance standards (e.g., ISAE 3000/3402), how to treat off-chain liabilities and derivatives, sampling expectations, and what qualifies as verifier independence. A simple public registry of attestors and methods would make weak attestations harder to hide and strong ones easier to recognise (FSB, 2023a, 2023b). To strengthen structural assurances more broadly—the backbone of institution-based trust—firms could also report a small set of outcomes on complaints and recourse, such as resolution times and compensation rates, which users can interpret at a glance (McKnight & Chervany, 2001).

Supervisors can further improve outcomes by testing, not guessing. “Evidence labs” and regulatory sandboxes that A/B-test incident summaries, risk labels, and solvency cards for comprehension and behavioural calibration would identify what actually works in practice; regulators could then publish templates that the market converges on (IOSCO, 2023). To avoid duplication and lower burden, these expectations should map directly to established frameworks—most notably the GOVERN function in NIST CSF 2.0 and ISO/IEC 27001:2022—so that one evidentiary set serves prudential, conduct, and consumer-protection aims simultaneously (NIST, 2024; ISO, 2022).

In short, the policy lever is not to mandate identical interfaces, but to make verifiable evidence the norm and to insist on legibility. If platforms can show consistent artefacts with known content and rhythm, users will be better able to judge whom to trust—without slowing responsible innovation (IOSCO, 2023; FSB, 2023a, 2023b; ESMA, 2025; EIOPA, 2025).

5.4. The Trust–Risk Capability Matrix

A practical way to translate the results into action is to place platforms on a simple 2×2: perceived trust on the vertical axis (low to high, as reported by users) and actual ICT risk capability on the horizontal axis (low to high, as evidenced by controls, testing, and resilience). The two axes capture the core finding of this study: trust moves when governance is both real and legible. Movement to the right reflects what the organization can demonstrably do, including secure development, disciplined custody, incident response, resilience testing, and third-party oversight. Movement upward reflects what users believe, shaped by transparency, communication, and education.

In the lower-left quadrant (low capability × low trust), the platform is at risk. Evidence is sparse or dated, incident communication is thin, custody and operations look immature, and change control appears ad hoc. Users are already discounting the platform, and the technical baseline

does not contradict them. Priorities here are defensive and concrete: establish incident runbooks with defined roles, timelines, and disclosure triggers; harden key management (for example, HSM or MPC with dual control); implement monitoring, alerting, and basic SLOs; and publish a minimum assurance pack that includes scope-clear SOC or ISO evidence, a short custody summary, and a live status page. Fast, candid disclosure and the first closed-loop post-mortem are early proofs of life. Clear exit signs from this quadrant include a fresh, scope-defined attestation, a timely post-mortem with remediation closed, and a reserve proof paired with explicit liabilities and a named independent attester.

In the upper-left quadrant (high capability $\times$ low trust), governance is strong but largely invisible. Audits, white papers, and runbooks exist, yet they are scattered, dense, or written only for experts. The survey's strongest pattern—communication quality shaping whether trust guides platform choice—points directly at this quadrant. The remedy is translation and curation rather than more control for its own sake: consolidate evidence in a single Trust Center; publish a dated status history with plain-language post-mortems; add concise risk nutrition labels that separate market, technical and operational, and counterparty risks; show remediation clocks and when they reach zero; and explain solvency signals in human terms (assets and liabilities, method, attester). Success looks like a visible rise in communication and transparency scores and a tighter link between trust in ICT management and stated platform choice, achieved without materially changing the underlying control set.

In the lower-right quadrant (low capability $\times$ high trust), the platform faces precarious over-trust. A brand halo or persuasive marketing has outpaced actual controls. Day to day may feel fine until it does not. Management should pause expansion and treat governance as the critical path: raise custody and change-management maturity, commission independent reviews, and bring messaging back to reality so expectations do not exceed what the system can safely deliver. Communication should remain candid, even if less rosy for a time. Progress needs to move right first (capability), then up (visibility), with tangible proofs such as closed audit findings, tested recovery procedures, and independent assurance for material claims.

In the upper-right quadrant (high capability $\times$ high trust), strong controls are matched by clear, routine communication, the resilient and legible state. Users anchor on evidence, not slogans. The task is to keep the cadence: rehearse and publish resilience tests, maintain attestation freshness, report misses as plainly as successes, and keep improving education for under-confident segments so gains are broad based rather than confined to power users. The main risk here is complacency: stale evidence and long-unupdated pages erode trust as surely as technical failure. Treat last updated dates as product SLAs, not decoration.

The matrix is useful because it suggests paths, not just labels. A safe rule of thumb is right, then up when capability is weak. No amount of marketing can erase governance debt. Conversely, where capability is already high (the more common case in this sample), up, then keep right is efficient: translate what exists, simplify the story, and continue strengthening the baseline in the background. In all quadrants, management should steer by a handful of leading indicators that are hard to game and easy to explain: time to post-mortem; the share of incidents with remediation milestones closed on time; attestation freshness and scope clarity; cadence and coverage of proof of reserves with liabilities; and education reach measured alongside comprehension and retention effects.

Finally, the matrix accommodates the heterogeneity seen in the data. Novices and lower-confidence users, more often women in this sample, benefit from plain labels, progressive disclosure, and default-safe settings (for example, withdrawal allow-lists on by default) to move upward on perceived trust. Experienced users tend to reward deeper evidence, such as full audit letters, method notes, and on-chain verifiers. Designing evidence for both audiences is what keeps platforms in the resilient and legible quadrant over time.

5.5. Study Limitations and Reliability Considerations

This study has several limitations that shape how the findings should be interpreted. First, the survey is cross-sectional and relies on self-report. As a result, the observed associations are correlational rather than causal and remain susceptible to common-method variance despite standard procedural safeguards (neutral wording, mixed item directions, attention checks) and diagnostic tests (a marker-variable approach and Harman's single-factor screen). In line with recent FinTech research, the discussion therefore uses cautious language (associated with, predicts, aligns with) rather than making causal claims (Amnas et al., 2023).

Measurement quality is also uneven across constructs. Four multi-item scales exhibit good to excellent internal consistency ($\alpha \approx .81-.92$). The two-item Transparency Expectations index performs adequately for a short scale ($\alpha = .672$), whereas the three-item Communication Views mini-scale is under-specified ($\alpha \approx .49$). The sizeable effects associated with communication should therefore be treated as provisional until replicated with a richer battery that captures the frequency, clarity, timeliness, and credibility of updates. Descriptive summaries treat 1–5 Likert responses as approximately interval for means and standard deviations; by contrast, inferential

analyses emphasize rank-robust statistics (for example, Spearman's ρ) and heteroscedasticity-consistent errors to respect ordinal properties (Hair et al., 2021).

Item-level missingness is not uniform. A small number of questions, most notably the incident-reporting statement, attracted materially fewer responses. Those estimates are flagged and should be read cautiously because pairwise deletion can bias results if responses are not missing at random. A focused re-fielding of thin items would improve precision and reduce the risk of selective non-response.

Sampling choices further limit generalizability. The frame is EEA-based and skews younger (18–34) with a sizeable novice segment. These features suit the research questions but may not transport to older or non-EEA populations without adjustment. Mode effects (panel versus community recruitment) and multihoming could also shape perceptions in ways not fully observed here. Robustness checks mitigate, but cannot eliminate, these concerns.

The qualitative strand is document-centric, which brings auditability and time stamps but inherits survivorship and presentation bias. Well-governed, or simply louder, firms tend to leave a richer paper trail, while opaque platforms may appear better or worse than they are. Mitigation rests on strict inclusion criteria (authenticity, clear as-of dates, scope clarity), triangulation with supervisory sources, and a transparent evidence rubric. The resulting documentary trust-signals index is a formative composite, meaning that its items cause the score rather than reflect a latent factor, so internal-consistency metrics are not applicable. Content validity instead rests on a published rubric, double-checks of binary extractions, and versioned archives.

Taken together, these constraints do not overturn the central pattern: legible governance and clear communication align most closely with trust-based platform choice. They do, however, argue for careful interpretation and replication. A natural next step is longitudinal and experimental work, including panel surveys around governance events, A/B tests of risk labels and post-incident messaging, and quasi-experiments that link documentary changes (for example, adding proof of liabilities to proof of reserves) to subsequent shifts in measured trust and behavior (Amnas et al., 2023; Hair et al., 2021).

5.6. Directions for Future Research

The most direct next step is to move beyond what users say they value and identify which concrete governance and communication choices actually shift trust and behavior. That requires designs that connect specific, time-stamped artifacts, such as audits, solvency proofs with liabilities, status histories, and plain-language labels, to measurable changes in perceptions and use.

A first priority is better measurement of communication quality. The present mini-scale captured an important signal but was too lean for high-stakes inference. Future work should specify a multi-item, multi-facet construct that covers the frequency, clarity, timeliness, and credibility of platform updates, and then validate it rigorously through cognitive testing, pilot work, confirmatory factor analysis, convergent and discriminant validity, and measurement invariance across languages and user segments. A short bank of items per facet (for example, three to four) would support a higher-order “communication quality” factor that generalizes across contexts (Hair et al., 2021).

A second priority is to test artifacts in the wild. Randomized A/B experiments can evaluate risk-label designs, the presence and depth of status-page history, and the content and timing of post-incident summaries on comprehension, perceived risk, and protective settings such as enabling MFA or withdrawal allow-lists. Quasi-experiments, including difference-in-differences and synthetic controls, can be anchored on time-stamped governance events such as publishing proof of reserves with proof of liabilities and a named independent attester, or reaching license and authorization milestones. Linking behavioral telemetry (retention, deposit and withdrawal flows, support burden) to repeated trust measures will show whether observed improvements persist beyond a single visit (Amnas et al., 2023).

Third, longitudinal designs should follow users through incident cycles. Event-aware panels that bracket a lifecycle (pre-event baseline → shock → remediation → recovery) would test whether institutional assurances, such as audits, authorizations, and disclosure SLAs, cushion dips in artifact trust and accelerate the normalization of usage. Such designs directly probe the spillovers and cross-layer dynamics proposed in layered trust models for FinTech (Devlin et al., 2025).

Fourth, individuals should be explicitly linked to documented platform signals in multi-level models. A formative, platform-level documentary index that captures, for example, PoR+PoL presence, attester independence, cadence of updates, custody disclosures, and incident post-mortems can be joined to each respondent’s primary platform and analyzed using mixed-effects regression or multilevel SEM. This enables tests of the pathway from structural assurances to

trusting beliefs to choice, while accounting for clustering by platform (and, in sensitivity checks, by country) and recognizing the index's formative nature.

Fifth, equity and heterogeneity should be treated as core questions rather than afterthoughts. Studies should oversample novices and lower self-efficacy groups, including women, who in this sample reported lower confidence and higher perceived risk, and should pre-register subgroup hypotheses. Experiments can then test whether “legibility lifts” (plain labels, progressive disclosure, visible status history) close confidence and trust gaps without harming expert users, looking explicitly for interaction effects rather than only averages (Amnas et al., 2023).

Sixth, policy design deserves prospective evaluation. Standardized assurance packs (PoR+PoL content and cadence, incident-disclosure SLAs, attestor identity and independence, scope-clear SOC/ISO evidence) and plain-language labels for custody and protocol classes should be tested for comprehension and behavioral calibration. In the EU, MiCA and DORA provide natural anchors and event windows for pre/post designs that assess whether visible compliance artifacts lift perceived regulatory support, trust, and continuance intention (IOSCO, 2023; EIOPA, 2025; ESMA, 2025).

Two cross-cutting practices would make these efforts more credible and cumulative. First, pre-registration with clear estimands and analysis plans to guard against researcher degrees of freedom. Second, open materials, including item banks, code, and anonymized artifact rubrics, so effects can be replicated and compared across platforms and jurisdictions. In short, the evidence so far supports a practical thesis: governance builds trust when it is both strong and legible. The research agenda should now identify, with experimental and longitudinal designs, which specific artifacts most efficiently convert structural assurance into lower perceived risk, higher trusting beliefs, and ultimately, trust-based platform choice (Amnas et al., 2023; Devlin et al., 2025; Hair et al., 2021; IOSCO, 2023; EIOPA, 2025; ESMA, 2025).

6. CONCLUSION

This thesis asked a simple question with complex consequences: what evidence, beyond slogans, must crypto platforms show to earn and keep user trust, and how does that trust shape behavior? Using documentary multiple case studies (Coinbase, Binance, Bitstamp), a cross-

sectional EEA survey, and independent benchmarks, the answer is clear: trust moves when governance is both strong and legible.

Users report high expectations for transparency and communication and a readiness to switch when those expectations are unmet, while baseline trust in security, data protection, and fund safety sits near the midpoint and is polarized. Perceived communication quality is the strongest unique correlate of whether trust guides platform choice. Self-efficacy rises with experience and travels with trust. In this EEA sample, women on average reported lower self-efficacy and higher perceived risk than men despite similar transparency expectations. Category-level risks (for example, policy shifts) are only weakly linked to platform-specific trust, suggesting that users separate exogenous uncertainty from provider-controlled safety.

The cases explain the mechanism. Artefacts that travel, such as status pages with dated post-mortems, scope-clear SOC/ISO attestations, proof of reserves paired with proof of liabilities (PoR+PoL) under a named, independent attestor, custody summaries (HSM/MPC), and visible change and incident routines, consistently convert back-office governance into structural assurances. Where such artefacts exist but are scattered or accessible only to experts, capability is real but legibility is low, and trust lags. Where they are absent or stale, skepticism is rational.

Contributions. Theoretically, the findings reinforce a layered view of trust in FinTech. Institution-based safeguards set the floor. Trust in technology rises when safeguards are made visible. Communication is the bridge from safeguards to intentions and choice. Methodologically, a document-first case approach, with explicit inclusion rules, hashing and time-stamping, and a formative documentary trust-signals index, is a viable substitute for interviews in disclosure-rich settings. Practically, the thesis specifies a shippable Governance Assurance Pack and a small trust KPI set: time to post-mortem, on-time remediation, attestation freshness and scope, PoR+PoL cadence, education reach and lift, and safe-configuration adoption.

Policy. Standardize evidence and insist on legibility. A lightweight, machine-readable schema for the assurance pack aligned to MiCA/DORA and mapped to NIST CSF 2.0 (GOVERN) and ISO/IEC 27001 would curb “trust theater” without freezing design. Guidance should clarify acceptable PoR/PoL practices and support tested, plain-language risk labels.

Managerial map. The Trust–Risk Capability Matrix summarizes the path: move right to raise real ICT-risk capability; move up to make that capability visible and rewarded. Where capability is weak, go “right, then up.” Where capability is strong, go “up, then keep right.”

Limits and next steps. The survey is cross-sectional and self-reported. One communication mini-scale is intentionally lean. The documentary strand faces survivorship and presentation bias despite safeguards. The EEA sample skews younger. Replication with event-aware panels, A/B tests (risk labels, post-incident summaries), and quasi-experiments around dated governance events (for example, adding liabilities to reserve proofs) is the logical next step.

Bottom line. In crypto markets, trust is earned where controls and candor meet. Strong engineering and disciplined operations shape behavior only when converted into clear, current, and checkable evidence that ordinary users can interpret. Publishing that evidence with rhythm and clarity, and aligning it to common schemas, moves the sector from reactive assurance to proactive, legible resilience.

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LIST OF FIGURES

Table 1. Sample composition	37
Table 2. Scale reliability and descriptives (1–5)	37
Table 3. Headline item statistics (selected; 1–5 Likert)	38
Table 4. Spearman correlations with the outcome	39
Table 5. Exploratory multivariate model (standardized OLS)	39
Table 6. Theme overview and typical evidence	40
Table 7. Means by crypto experience (1–5)	41
Table 8. Experience gradient and tests	42
Table 9. Means by gender (1–5)	42
Table 10. Gender tests (Kruskal–Wallis)	42
Table 11. Age tests (Kruskal–Wallis)	43
Table 12. Construct-level triangulation (survey ↔ documentary/secondary)	44